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MTH 102 — Past Questions & Materials
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Course: MTH 102
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 100

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Paper 1 of 4 — Mth 102 2023/2024 past question

Level 100 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUM, DELTA STATE.
COLLEGE OF SCIENCE, DEPARTMENT OF MATHEMATICS.
MTH 102/121 EXAMINATIONS SECOND SEMESTER, 2023/2024 SESSION,
TIME ALLOWED: 1HR 15 MINS Instruction: Answer all questions TYPE A

- The derivative of a constant function/term is? (A) 0 (B) $2x$ (C) x^2 (D) $4x^2$
- What is the second derivative of the function $y = 4x + 1$? (A) 0 (B) 4 (C) 1 (D) 2
- Differentiating the function $y = (4x - 6)^2$ gives? (A) $4(4x - 6)$ (B) $8(4x - 6)$ (C) $(4x - 6)$ (D) $2(4x - 6)$
- The hyperbolic function $\sinh x$ is defined as? (A) $\frac{e^x - e^{-x}}{2}$ (B) $\frac{e^x + e^{-x}}{2}$ (C) $\frac{e^x - e^{-x}}{4}$ (D) $\frac{e^x + e^{-x}}{4}$
- $\tanh^2 x - \operatorname{sech}^2 x$ is equivalent to? (A) 1 (B) 2 (C) 4 (D) 10
- Evaluate $\frac{dx}{9+4x^2}$ (A) $\frac{1}{3} \tan^{-1} 2x + c$ (B) $\frac{1}{6} \tan^{-1} 2x + c$ (C) $\frac{1}{2} \sec^{-1} 2x + c$ (D) $\frac{1}{3} \sec^{-1} 2x + c$
- Simplify $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ (A) -6 (B) -7 (C) 7 (D) 6
- Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x^2 - 6x + 8}$ (A) $-\frac{1}{2}$ (B) $-\frac{1}{3}$ (C) $-\frac{1}{4}$ (D) $-\frac{1}{5}$
- Simplify $\lim_{x \rightarrow \infty} \frac{1+x^2}{1-x^2}$ (A) -3 (B) -1 (C) -2 (D) -4
- Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 4x + 1}{3x^2 - x + 5}$ (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $-\frac{2}{3}$ (D) $-\frac{1}{3}$
- Find the volume of the solid of revolution between the curve $y = (x + 2)^2$ in the interval of $x = 0$ and $x = 1$ leaving your answer in terms of π (A) $\frac{12\pi}{5}$ (B) $\frac{19\pi}{3}$ (C) $\frac{12\pi}{9}$ (D) $\frac{19\pi}{5}$
- Calculate the volume of the curve $y = (x^2 - 1)^2$ between the interval -1 and 1 (A) $\frac{16\pi}{5}$ (B) $\frac{16\pi}{15}$ (C) $\frac{16\pi}{12}$ (D) $\frac{16\pi}{13}$
- Calculate the area of the segment cutting across the curve $y = x(3 - x)$ in the interval of $x = 0$ and $x = 2$ (A) $\frac{10}{4}$ (B) $\frac{10}{3}$ (C) $\frac{10}{6}$ (D) $\frac{10}{8}$
- The property of definite integral defined as $\int_a^b k f(x) dx = k \int_a^b f(x) dx$ and $\int_a^b \{f(x) + g(x)\} dx = \int_a^b f(x) dx + \int_a^b g(x) dx$ is called (A) Additivity of adjacent integral (B) Linearity of definite integral (C) reversing the integration (D) Multiplicative property definite integral.
- Find the domain of the function $g(x) = 4x + 1$ (A) $-\infty < x < 4$ (B) $-4 < x < \infty$ (C) $-\infty < x < \infty$ (D) $-\infty < x < 3$
- The composition of two functions $f(x)$ and $g(x)$ is defined by (A) $(f \circ g)(x) = f(g(x))$ (B) $(f \circ g)(x) = \frac{f}{g}(g(x))$ (C) $(f \circ g)(x) = g(x) - f(x)$ (D) $(f \circ g)(x) = \frac{(f-g)(x)}{4}$
- Given that $f(x) = 1 - x$ and $g(x) = x - 1$. Find $(f \circ g)(x)$ (A) $2 + x$ (B) $1 + x$ (C) $1 - x$ (D) $2 - x$
- Given that $h(x) = x^2$ and $g(x) = 2 + x$. Find $(h \circ g)(x)$ (A) $x^2 + 4x - 4$ (B) $x^2 + 4x + 4$ (C) $x^2 - 4x + 4$ (D) $x^2 - 4$
- Evaluate $\lim_{x \rightarrow 1} \frac{2-x}{1-4x}$ (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$
- Given the exponential function $y = e^{2x^2 - 2}$, find its derivative with respect to x (A) $4e^{2x^2 - 2}$ (B) $4e^{2x - 2}$ (C) $4xe^{2x^2 - 2}$ (D) $4e^{2x - 2}$
- Differentiate the function $y = (5x - 2)^3$ (A) $15x(5x - 2)^3$ (B) $15(5x - 2)^3$ (C) $15(5x - 2)$ (D) $15(5x - 2)^2$
- What is the derivative of the function $y = 3x^2 e^{2x}$ (A) $9xe^{2x}(x + 1)$ (B) $6e^{2x}(x + 1)$ (C) $6xe^{2x}(x + 1)$ (D) $9e^{2x}(x + 1)$
- Differentiating the function $y = x^3 \sin x$ gives? (A) $x^3(\cos x + 3 \sin x)$ (B) $x(\cos x + 3 \sin x)$ (C) $x^2(\cos x + 3 \sin x)$ (D) $x^3(\cos x + 3 \sin x)$
- What differential techniques is suitable to solve the function $y = \tan x e^{-2x}$ (A) product rule (B) Implicit rule (C) Quotient rule (D) chain rule.
- Given the implicit function $x^5 + xy - y^4 = 0$, find $\frac{dy}{dx}$ (A) $\frac{y+3x^2}{x-4y^3}$ (B) $\frac{y-3x^2}{x+4y^3}$ (C) $\frac{-y+3x^2}{x-4y^3}$ (D) $\frac{-y-3x^2}{x-4y^3}$
- Find $\frac{dy}{dx}$ from the function $x^2 - y^2 + x^3 + 6 = x^2 y^4$ (A) $\frac{3x+2x^2-2x^4}{2y+4x^2y^3}$ (B) $\frac{2x+3x^2-2xy^4}{-2y-4x^2y^4}$ (C) $\frac{5x-3x^2-4xy^4}{2y+4x^2y^3}$ (D) $\frac{2x+3x^2-2xy^4}{2y+4x^2y^3}$
- At what point is the second derivative of a function less than zero? (A) stationary point (B) inflexion point (C) maximum point (D) minimum point.
- What is the stationary point of the function $y = 2x^3 - \frac{1}{2}x^2 - x$? (A) $\frac{1}{2}$ and $-\frac{1}{3}$ (B) $\frac{1}{3}$ and $-\frac{1}{3}$ (C) $\frac{2}{3}$ and $-\frac{3}{2}$ (D) $\frac{2}{5}$ and $-\frac{5}{2}$
- Given that function $y = x^2 - 4x$, what point does the second derivative of the suggests? (A) minimum point (B) maximum point (C) stationary point (D) inflexion point
- Given that the volume of a cylinder is $V = \pi r^2 h$, if the volume V is differentiated with respect to height h , what does the derivative suggests? (A) Area of a cone (B) circumference of a circle (C) Area of circle (D) volume of a cylinder.
- Differentiate Z with respect to x from the function $Z = 2xy + x^2 + y^2$ (A) $2y + 2x$ (B) $2xy + 2x$ (C) $2x^2 + 2y$ (D) $2xy + y^2$
- Given the function $u = (4x - 2y)(3x + 5y)$, differentiate u with respect to y (A) $12x^2 - 26y^2$ (B) $-20x - 20y$ (C) $12x^2 - 10y^2$ (D) $26x - 20y$
- Differentiate u with respect to x from the function $(2x - 4y)(4x + 7y)$ (A) $16xy + 2y$ (B) $16x^2 - 2y^2$ (C) $16x - 2y$ (D) $16y - 2x^2$
- Given that $\frac{dy}{dx} = 3x^2 + 4$, perform the reverse process to obtain the original function y (A) $y = x^3 + 4x^2 + C$ (B) $y = 3x^3 + 4x + C$ (C) $y = 3x^2 + 4x + C$ (D) $y = x^3 + 4x + C$
- Evaluate the integral $\int 3x^2 dx$ (A) $2x^2 + C$ (B) $3x^2 + C$ (C) $3x + C$ (D) $3x^{-3} + C$
- Evaluate the integral $\int (5x - 4)^4 dx$ (A) $\frac{(5x-4)^4}{20} + C$ (B) $\frac{(5x-4)^4}{4} + C$ (C) $\frac{(5x-4)^5}{25} + C$ (D) $\frac{(5x-4)^4}{9} + C$

37. Evaluate $\int \frac{x^3}{\sqrt{1-x^4}} dx$ (A) $\frac{(1-x^4)^2}{2x} + C$ (B) $\frac{(1-x^4)^2}{2} + C$ (C) $\frac{(1-x^4)^2}{2} + C$ (D) $\frac{(1-x^4)^2}{2x} + C$
38. Evaluate $\int x^4 \ln x dx$ (A) $\frac{x^4}{4} \ln x - \frac{1}{4}$ (B) $\frac{x^4}{4} \left(\ln x - \frac{1}{4} \right)$ (C) $\frac{x^5}{5} \ln x - \frac{1}{5}$ (D) $\frac{x^5}{5} \left(\ln x - \frac{1}{5} \right)$
39. Evaluate the integral $\int \frac{7x+1}{(3x-1)(2x+3)} dx$ using the partial fraction approach (A) $\frac{2}{11} \ln(3x-1) + \frac{3}{11} \ln(2x+3)$ (B) $\frac{10}{9} \ln(3x-1) + \frac{1}{9} \ln(2x+3)$ (C) $\frac{10}{11} \ln(3x-1) + \frac{19}{11} \ln(2x+3)$ (D) $\frac{5}{9} \ln(3x-1) + \frac{2}{9} \ln(2x+3)$
40. Evaluate the integral $\int \frac{4x^3-1}{x^4-x+2} dx$ (A) $\ln(x^4-x+2) + C$ (B) $\ln(x^4-x+2)^4 + C$ (C) $\ln(4x^3-x+2) + C$ (D) $\ln(x^4-x+2)^3 + C$
41. Evaluate $\int \frac{dx}{2x-1}$ (A) $\frac{1}{2} \ln(2x-1)^2 + C$ (B) $\frac{1}{2} \ln(2x^2-1)^2 + C$ (C) $\frac{1}{2} \ln(2x-1) + C$ (D) $\frac{1}{3} \ln(2x-1) + C$
42. Evaluate the integral $\int e^{(3x+1)} dx$ (A) $\frac{e^{(3x+1)}}{3} + C$ (B) $-\frac{e^{(3x+1)}}{3x} + C$ (C) $\frac{e^{(3x+1)}}{2x} + C$ (D) $\frac{e^{(3x+1)}}{2} + C$
43. Evaluate the integral $\int \frac{\sin x}{\cos x} dx$ (A) $\ln|\tan x + \operatorname{cosec} x| + C$ (B) $\ln|\operatorname{cosec} x + \sec x| + C$ (C) $\ln|\sec x + \tan x| + C$ (D) $\ln|\tan x - \operatorname{cosec} x| + C$
44. Evaluate the integral $\int \frac{dx}{9+4x^2}$ (A) $\frac{1}{3} \tan^{-1} \frac{2x}{3} + C$ (B) $\frac{1}{2} \sec^{-1} \frac{2x}{3} + C$ (C) $\frac{1}{6} \tan^{-1} \frac{2x}{3} + C$ (D) $\frac{1}{4} \sec^{-1} \frac{2x}{3} + C$
45. Evaluate the integral $\int \frac{x+2}{\sqrt{4-x^2}} dx$ (A) $-\sqrt{4-x^2} + 2\sin^{-1} \frac{x}{2} + C$ (B) $\sqrt{4-x^2} + 2\sin^{-1} \frac{x}{2} + C$ (C) $-\sqrt{4-x^2} + \tan^{-1} \frac{x}{2} + C$ (D) $\sqrt{4-x^2} + 2\tan^{-1} \frac{x}{2} + C$
46. Evaluate the integral $\int \frac{3x^2-2}{x^2+4} dx$ (A) $\frac{3x}{2} + 6 \ln(x-4) - \sin \frac{x}{2} + C$ (B) $\frac{3x}{2} + 6 \ln(x+4) + \tan \frac{x}{2} + C$ (C) $\frac{3x}{2} + 6 \ln(x-4) - \sin^{-1} \frac{x}{2} + C$ (D) $\frac{3x}{2} + 6 \ln(x-4) - \tan^{-1} \frac{x}{2} + C$
47. Find the derivative of $y = \sin(4x-1)$ (A) $4\sin(4x-1)$ (B) $4\cos(4x-1)$ (C) $4x\cos(4x-1)$ (D) $4x\sin(4x-1)$
48. Given the integral $\int \frac{x+1}{x^2+2x-5} dx$, obtain its derivative the reverse process. (A) $\frac{1}{2} \ln(x^2+2x-5)$ (B) $\frac{1}{2x} \ln(2x^2+2x-5)$ (C) $\ln(x^2+2x-5)$ (D) $\ln(x^2+2x-5)$
49. Evaluate the integral $\int \frac{e^{3x}}{e^{3x}-1} dx$ (A) $\frac{1}{2} \ln(2x-1) + C$ (B) $\ln(2x-1) + C$ (C) $\frac{1}{3} \ln(3x-1) + C$ (D) $\ln(3x-1) + C$
50. What is the value of the integral $\int_1^4 x^2 dx$ (A) 21 (B) 23 (C) 25 (D) 27
51. Evaluate the integral $\int \cos x dx$ within the interval of 0 and $\frac{\pi}{2}$ (A) 3 (B) 0 (C) 1 (D) 5
52. What is the value of the integral $\int (x^2 - 3x + 1) dx$ when it is evaluated with the interval of -2 and 4? (A) 64 (B) 12 (C) 16 (D) 14

Use the information below to answer question 53, 54 and 55 respectively

Given that the curve $y = x(x-1)(x-2)$ is within the interval of the areas A and B such that the area of A is with the interval 0 and 1 above the x -axis while the area of B is with the interval of 1 and 2 below the x -axis

53. Evaluate the area A of the curve between the interval of 0 and 1 (A) 4 (B) $\frac{1}{4}$ (C) 2 (D) $\frac{1}{2}$
54. Evaluate the area B of the curve between the interval of 1 and 2 (A) 8 (B) $-\frac{1}{4}$ (C) 4 (D) $\frac{1}{16}$
55. Evaluate the from A to B (A) 0 (B) 12 (C) 4 (D) 1
56. Given that the derivative of a function is $\frac{dy}{dx} = 2x^2 + 3$, what is the value of the constant of integration if it is evaluated at $x = 2$ and $y = 2$? (A) $9\frac{1}{3}$ (B) $9\frac{2}{3}$ (C) $9\frac{3}{3}$ (D) $9\frac{4}{3}$
57. Find the area bounded by the curve $y = x^2 + x + 4$ within the interval of $x = 1$ and $x = 3$ (A) $20\frac{3}{2}$ (B) $20\frac{1}{2}$ (C) $20\frac{2}{3}$ (D) $20\frac{1}{3}$
58. Calculate the value of the definite integral $\int_1^6 (4x - x^2) dx$ (A) $2\frac{1}{2}$ (B) $2\frac{3}{2}$ (C) $1\frac{2}{3}$ (D) $1\frac{1}{2}$
59. Find the derivative of the function $y = 2x^3 + x^2 + 3x - 10$ (A) $6x^2 + 2x - 3x$ (B) $6x^3 - 2x - 3$ (C) $6x^2 - 2x - 3$ (D) $6x^2 + 2x + 3$
60. A function $F: X \rightarrow Y$ is called a constant function if (A) $f(x) = x$ (B) $f(x) = y$ (C) $f(x) = b$ (D) $f(x) = x^2$
61. A function $F: X \rightarrow Y$ is called an identity function if (A) $f(x) = 4y$ (B) $f(x) = 4x$ (C) $f(x) = y$ (D) $f(x) = x$
62. What conic section does the equation above represent? (A) Parabola (B) Circle (C) Ellipse (D) Hyperbola
63. A function is said to be linear if it is of the form (A) $ax + b$ (B) $ax^2 + b$ (C) $a + bx^2$ (D) $a - b$
64. Evaluate $h(2)$, if $h(x) = 1 - x^2$ (A) 3 (B) 4 (C) -4 (D) -3
65. Find the range of the function $f(x) = 2x + 1$ (A) $-\infty < y < 2$ (B) $-2 < y < \infty$ (C) $-\infty < y < \infty$ (D) $-\infty < y < 3$

Use the information below to attempt question 66 to 68

Suppose $f(x) = \begin{cases} -x^2, & \text{for } x < -1 \\ \sqrt[3]{x^3}, & \text{for } x \geq -1 \end{cases}$

66. Evaluate $f(-3)$; (A) undefined (B) -27 (C) -9 (D) 9
67. Evaluate $f(0)$; (A) 0 (B) -1 (C) 1 (D) $\frac{1}{3}$
68. Evaluate $f\left(\frac{1}{2}\right)$; (A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) $\frac{3}{4}$
69. $\frac{d}{dx} (1 + 2x^2)^{-1}$. (A) $-(1 + 2x^2)^{-2}$ (B) $\frac{-(1+4x)}{(1+2x^2)^2}$ (C) $\frac{1+4x}{1+2x^2}$ (D) $\frac{-4x}{(1+2x^2)^2}$
70. Find the derivative of the function $y = e^{\sin 2x}$. (A) $2\cos 2x e^{\sin 2x}$ (B) $-2\sin 2x e^{\cos 2x}$ (C) $\frac{-2\sin 2x}{e^{\cos 2x}}$ (D) $\ln e^{\cos 2x}$

Paper 2 of 4 — MTH 102 Data Analysis PQ 2014-2020

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2019/2020

Course Code: MTH 112

Title: Data Analysis

Level: 100

Duration: 2 hrs

Name: EU WABOND CEALWA Matric No: CD51595212019**TYPE D**

- If the var (X) = 15. Find the cov (aX, X) if X is a random variable of a population and a = 2 (a) 15 (b) 225 (c) 30 (d) 15a
- What is the value of x if the $2x - 1$ is the sectorial angle of $1/12$ with a pie-chart whole cumulative is 12 (a) 2 (b) 1 (c) $1\frac{1}{2}$ (d) 12
- Find the var (X-Y) if X and Y are independently and identically distributed if the var (X), var (Y) and cov(X, Y) are $2\frac{1}{2}$, 4, and $\frac{1}{2}$ respectively. (a) 6 (b) 7 (c) $6\frac{1}{2}$ (d) 3
- When frequency data and count data are represented by counts or number of observations in each group of observation in a population, they are referred to as --- data (a) categorical (b) measurement (c) quantitative (d) parameter
- assigns numbers to object of a sample in an experiment. (a) measurement (b) parameter (c) statistic (d) nominal scale
- Find the mean of the distribution 66, 75, 72, 71, 55, 56, 72, 93, 72, 72, 73, 73, 91, 66, 71, 56, 59. (a) 66.4 (b) 72.00 (c) 66.3 (d) 70
- Which of the following represent the characteristics of a good questionnaire? (a) Tele-guiding questions (b) Long and explanatory questions (c) Honest and vogue questions (d) Unambiguous

The table below represents the examination scores of 100 levels students in MTH 112 in 2013/2014 session with mean 18.38. Use the table to answer questions 21 to 23

x	6-10	11-15	16-20	21-25	26-30
f	7	7	11	10	6

- Find the variance of the distribution (a) 6.00 (b) 18.38 (c) 41.52 (d) 6.40
- Find the value of y in the distribution (a) 6.0 (b) 18.00 (c) 41.00 (d) 8.00
- What is the median of the distribution? (a) 6.0 (b) 18.3 (c) 41.00 (d) 21.00
- Find the mode of 66, 75, 72, 71, 55, 56, 72, 93, 72, 72, 73, 73, 91, 66, 71, 56, 59, 56, 72, 93, 72, 72, 73, 73 (a) 66 (b) 72 (c) 56 (d) 73

Use the information to answer questions 12 - 15

x	1.2	1.3	1.4	2.4	0.8	3.4	0.2	1
y	5	6	1	2.9	4.2	k	5	3

- What is the value of k if the slope is 1.327? (a) 7 (b) 3 (c) 10 (d) 8
- Find the intercept if $y = a + bx$ is the regression equation (a) 3 (b) 7 (c) 8 (d) 10
- Which of the equation represents the regression equation? (a) $y = 3 + 1.3x$ (b) $y = 7 + 1.8x$ (c) $y = 3 - 1.8x$ (d) $y = 4.2 + 8x$
- What is the mean of the variable y? (a) 4.64 (b) 1.46 (c) 10.00 (d) 1.00
- What is the value of y if the mean of the number 13, $2y + 1$, 0, $5y - 1$, and 11 is 9? (a) 1.7 (b) 3.0 (c) 4.0 (d) 7.0
- The mean and the median of a symmetric distribution is said to be equal when they have -- distribution? (a) unimodal (b) bimodal (c) multimodal (d) mode
- Acceptance of the null hypothesis when it is false leads to ----- error. (A) type I (B) two-tailed (C) one-tailed (d) type II
- The first step in hypothesis testing is the determination of ---- (a) H_0 and H_1 (b) critical region (c) calculated value of the test statistic (d) decision
- If $n = 148$, $\bar{x} = -7.1$, $\sigma = 1.2$. A 95% confidence interval for μ is ----- (a) (6.91, 7.29) (b) (-7.29, -6.91) (c) (-7.29, 6.91) (d) (-6.91, -7.29)
- If $n = 16$, $\bar{x} = -7.1$, $S = 1.1$. A 95% confidence interval for μ is ----- (a) (6.51, 7.69) (b) (6.69, 7.51) (c) (-6.51, 7.69) (d) (-7.69, -6.51)
- The staff strength of Jom-oil Company is shown below. Find the median. (a) 44.07 (b) 42.00 (c) 77.00 (d) 45.00

Staff strength	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
No. of company	1	5	10	19	42	10	6	4	2	1

- The time taken in seconds by 100 different chemical substances to melt when subjected to a particular temporary condition are given below. Find the mode (a) 6.145 (b) 6.801 (c) 6.965 (d) 35

Time (sec)	4.51 - 5.32	5.33 - 6.14	6.15 - 6.96	6.97 - 7.78	7.79 - 8.60	8.61 - 9.42
frequency	15	7	35	28	10	5

- Find the probability of the class interval 30 - 39 in the table below (a) 1 (b) $3/10$ (c) $1/5$ (d) $1/12$

x	0-9	10-19	20-29	30-39	40-49	50-59
y	10	2	4	5	25	14

- What is the standard deviation of the distribution for an average of 48.28? (a) 366.96 (b) 18.96 (c) 359.63 (d) 19.16

X	25	29	y + 3	59	70
f	7	9	8	6	20

- Find the value of y for an average of 48.28 in the table below

X	y + 3	70	25	59	29
f	8	20	7	6	9

(a) 38 (b) 50 (c) 40 (d) 25

- Given the table below for 100 level 50 test scripts

11-15	16-20	21-25	26-30	31-35	36-40
6	5	x	12	9	5

If the class intervals are represented in a pie chart, what is the sectorial angle of the class interval 26-30? (a) 43.2 (b) 86.4 (c) 93.6 (d) 64.8

- Find the class size of this distribution

x	26-30	31-35	36-40	41-45
frequency	p	y	10	21

(a) 15.5 (b) 5.0 (c) 25.6 (d) 30.5

29. The normal distribution is represented by (a) $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ (b) $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu)^2}$ (c) $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ (d) $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$
- Use the information below to answer question 30 and 31
- | | | | | | |
|---|-------|-------|-------|-------|-------|
| x | 15-19 | 20-24 | 25-29 | 30-34 | 35-40 |
| f | 3 | 4 | 5 | 1 | |
30. Find the mean of the distribution (a) 27.50 (b) 26.25 (c) 7.00 (d) 30.00
31. Find the skewness (a) 0.40 (b) 2.25 (c) 0.22 (d) -0.40
32. The table below shows a frequency distribution for grouped data. Calculate the mode. (a) 24.75 (b) 22.37 (c) 27.73 (d) 27.50
- | | | | | | | |
|-----------|-----|------|-------|-------|-------|--------|
| frequency | 0-5 | 5-15 | 15-25 | 25-35 | 35-45 | 45-100 |
| x | 2 | 6 | 20 | 15 | 8 | 4 |
33. Find the sum of the deviation from the mean of the distribution 4, 6, 2, 8, 10, 3, 6, 4, 7, 10: (a) 22 (b) 6 (c) 0 (d) 7.8
34. _____ is a numerical value that summarizes population (a) Parameter (b) Statistic (c) Inferential statistic (d) Descriptive statistic
- Use the following information to answer questions 35 to 37.
- A random sample of 28 FUPRE students read a mean of 1.2 books with a standard deviation of 2.14 books in a month.
35. At 0.05 level of significance, determine the calculated value of the statistic for testing the claim that the mean number of books read per month is less than 2 (a) -0.80 (b) 0.80 (c) 1.98 (d) -1.98
36. The table value for this test is (a) 1.96 (b) 1.65 (c) 2.252 (d) 1.703
37. The decision for the hypothesis is (a) Reserve judgment (b) Accept H_0 (c) Cannot reject H_0 (d) Reject H_0
38. The variance of a set of data is 40, what is the variance if each element of the data is divided by 2 (a) 40 (b) 10 (c) 160 (d) 80
39. The variance of a set of data is 30, what is the variance if 2 is subtracted from each element of the data (a) 60 (b) 30 (c) 90 (d) 12
40. If $n = 25$, $\bar{x} = 15.15$, $S = 1.5$. A 95% confidence interval for μ is ---- (a) (14.56, 15.74) (b) (14.53, 15.77) (c) (6.51, 7.69) (d) (16.69, 17.51)
41. A company claims that its bag of rice weight 50kg. To test if the actual weight is less than the company claim, samples were selected and hypothesis testing carried out. What type of test is this? (a) type I (b) type II (c) two-tailed (d) one-tailed
42. H_0 for testing hypothesis using normal distribution is rejected if (a) $Z_{cal} < Z_{tab}$ (b) $P - value < \alpha$ (c) $Z_{cal} \leq Z_{tab}$ (d) $P - value > \alpha$
43. If $\alpha = 7.5\%$, what is the table value for a two-tailed test using normal distribution (a) 1.98 (b) 1.64 (c) 1.78 (d) 1.76
44. Determine the P-value for a one-tailed test if the calculated value of the normal distribution statistic is -1.72 (a) 0.0427 (b) 1.0962 (c) 0.9573 (d) 1.0952
45. The time taken by a typist to complete a task on a machine is normally distributed with $\sigma = 2.5$ mins. In a sample of 25 tasks, $\bar{x} = 25.5$ mins was found. Construct a 95% confidence interval for μ (a) (25.64, 26.96) (b) (24.52, 26.48) (c) (21.64, 26.96) (d) (24.52, 26.75)
46. The 99% confidence interval for μ given a data set with $n = 36$, $\bar{x} = 15.5$, $S = 0.75$ is (a) (13.50, 15.58) (b) (15.50, 15.58) (c) (15.18, 15.82) (d) (15.40, 15.50)
- Use the following information to answer questions 47 to 49. A 95% confidence interval reported for the parameter μ is 5.27 to 14.73. The interval is based on 5 observations constructed using t-distribution.
47. What is the table value of the t-distribution? (a) 2.131 (b) 2.776 (c) 2.582 (d) 1.642
48. What is the value of the standard error of mean ($s.e(\bar{x})$)? (a) 6.80 (b) 6.48 (c) 1.70 (d) 4.50
49. What is the mean of the unknown data set? (a) 10.00 (b) 11.50 (c) 12.50 (d) 20.50
50. Find the cov (20, a) if a is a constant of variation whose probability of occurrence is one. (a) 20 (b) 400a (c) 0a (d) a
- Use the following information to answer questions 51 to 53. The mean score in an examination from a random sample of 25 students is 47.50 with $S = 2.50$.
51. The calculated value of the statistic for testing the claim that the mean score is higher than 45 at $\alpha = 1\%$ (a) 2.81 (b) 2.58 (c) 5.64 (d) 5.00
52. The table value for the test is (a) 1.96 (b) 2.49 (c) 1.64 (d) 2.58
53. The hypothesis decision is (a) Reserve judgment (b) Accept H_0 (c) Reject H_0 (d) None of the above
54. The variance of a set of data represented by Y is 1.50 then the variance of $(Y - 2)$ is (a) 6.0 (b) 1.5 (c) 3.0 (d) 4.5
55. X represents a set of data with variance 5.0, the variance of $\left(\frac{X}{2}\right)$ is (a) 1.25 (b) 2.50 (c) 5.0 (d) 7.5
56. Let $\{(1,5), (3,4), (m,2), (4,5), (5,1), (1,3), (1,4), (2,5), (3,5), (0,6)\}$ be points on the Euclidean plane with equal means. Find the sample variance of x. (a) 20 (b) 19 (c) 34 (d) 6
57. Find the value of $\text{var}(X - Y)$ if the $\text{var}(X)$, $\text{var}(Y)$ and $\text{cov}(X, Y)$ are $2\frac{1}{2}$, 4, and $\frac{1}{2}$ respectively. (a) 7 (b) $6\frac{1}{2}$ (c) 3 (d) 6
58. The test records for mathematics and computer students are presented in pair respectively (1,2), (3,3), (3,5), (5,4), (5,6), (6,5), (8,7), (9,8). Find the cov (X, Y). (a) 3.50 (b) 6.25 (c) 4.24 (d) 5.00
59. Find the coefficient of variation for computer students of test records for Mathematics and Computer students (1,2), (3,3), (3,5), (5,4), (5,6), (6,5), (8,7), (9,8). (a) 50 (b) 70 (c) 37 (d) 125
60. Academic class of degree (First, Second, Third) is an example of ----- scale. (a) ordinal (b) nominal (c) interval (d) ratio
- Use the following information to answer questions 61 to 65. A random sample of 36 Shoprite customers spent a mean amount of 450.50 naira and standard deviation of 100.50 naira.
61. The standard error of means for this sample is (a) 6.75 (b) 1.96 (c) 1.65 (d) 16.75
62. The calculated value of the statistic for testing the claim that the mean amount spent is less than 400 at 0.05 level of significance is (a) 1.03 (b) 3.01 (c) 3.50 (d) 1.11
63. The table value for this test is (a) 1.96 (b) 2.33 (c) 1.65 (d) 2.58
64. The P-value for this test is (a) 0.0013 (b) 2.0313 (c) 0.6523 (d) 2.5823
65. The hypothesis decision is (a) Reserve judgment (b) Reject H_0 (c) Accept H_0 (d) None of the above
- Use the following information to answer questions 66 to 70. The following are the test scores of students 30, 20, 16, 10, 24.
66. The unbiased estimate of the population standard deviation is (a) 6.81 (b) 11.60 (c) 3.81 (d) 7.62
67. The standard error of mean for the sample is (a) 6.80 (b) 3.05 (c) 3.41 (d) 4.50
68. The calculated value of the statistic for testing the claim that the sample mean is not significantly different from 22 at $\alpha = 5\%$ is (a) -0.59 (b) 3.01 (c) -3.50 (d) 0.59
69. The critical value used for the test in question 33 is (a) 1.96 (b) 2.58 (c) 2.78 (d) 1.64
70. The hypothesis decision is (a) Reject H_0 (b) Accept H_0 (c) Reserve judgement (d) None of the above

$$1.) \text{Cov}(ax, by) = ab \text{Cov}(x, y)$$

$$\text{Cov}(ax, x) = a \text{Cov}(x, x)$$

$$\text{but } \text{Cov}(x, x) = \text{Var}(x)$$

{ Since x is a random variable, $n=1$ } where $\text{Var}(x) = \frac{\text{Var}(x)}{n}$

$$\therefore a \text{Cov}(x, x) = a \times \text{Var}(x) = a \times 15 = 2 \times 15 = 30 \quad (C)$$

$$2.) \text{Cumulative} = 12$$

$$x = ?$$

$$\frac{2x-1}{12} = \frac{1}{12}$$

$$\frac{1}{12}(2x-1) = \frac{1}{12}$$

$$2x-1 = 1$$

$$2x = 2$$

$$x = 1 \quad (B)$$

$$3.) \text{Var}(x-y) = \text{Var}(x) + \text{Var}(y) - 2\text{Cov}(x, y)$$

$$\text{Var}(x) = 2\frac{1}{2} = \frac{5}{2}$$

$$\text{Var}(y) = 4$$

$$\text{Cov}(x, y) = \frac{1}{2}$$

$$\text{Var}(x-y) = \frac{5}{2} + 4 - 1$$

$$\text{Var}(x-y) = 5.5$$

$$4.) (A)$$

$$5.) (A)$$

$$6. \text{REARRANGE} = 55, 56, 56, 59, 66, 66, 71, 71, 72, 72, 72, 72, 73, 73, 75, 91, 93$$

$$\bar{x} = \frac{\sum x}{n} = \frac{1193}{17} = 70.18$$

$$7.) (D)$$

8. x(classes)	frequency	mid point	mf	m ² f	c f
6-10	4	8	32	256	5.78
11-15	7	13	91	1183	12.78
16-20	11	18	198	3564	23.78
21-25	10	23	230	5290	33.78
26-30	6	28	168	4704	37.78

$$\text{mean } \bar{x} = 18.38$$

$$\bar{x} = \frac{\sum mf}{\sum f} = \frac{687 + 84}{34 + 4}$$

$$18.38 = \frac{687 + 84}{34 + 4}$$

$$687 + 84 = 18.38 (34 + 4)$$

$$687 + 84 = 624.72 + 18.384$$

$$62.08 = 10.384$$

$$4 = 5.78$$

$$S^2 = \frac{\sum m^2 f}{N} - \left(\frac{\sum mf}{N} \right)^2$$

$$S^2 = \frac{14741 + 644}{34 + 4} - \left(\frac{687 + 84}{34 + 4} \right)^2$$

$$S^2 = \frac{14741 + 64(5.78)}{34 + 5.78} - \left(\frac{687 + 8(5.78)}{34 + 5.78} \right)^2$$

$$S^2 = \frac{15123.72 - 337.83}{37.78} = 369.83$$

9. (A)

$$10. \text{Median} = L + \frac{\frac{N}{2} - F}{f} \times h$$

$$\frac{N}{2} = \frac{37.98}{2} = 17.99$$

$$\text{median class} = 16 - 20$$

$$\begin{aligned} \text{Median} &= 16 + \left(\frac{\frac{37.98}{2} - 12.98}{11} \right) \times 4 \\ &= 16 + \left(\frac{7.01}{11} \right) \times 4 \\ &= 16 + 2.55 \\ &= 18.55. \quad (B) \end{aligned}$$

11. (B)

12	n	x	y	x ²	y ²	xy
1	1	1.2	5	1.44	25	6
2	1	1.3	6	1.69	36	7.8
3	1	1.4	1	1.96	1	1.4
4	1	2.4	2.9	5.76	8.41	6.96
5	1	0.8	4.2	0.64	17.64	3.36
6	1	3.4	k	11.56	k ²	3.4k
7	1	0.2	5	0.04	25	1
8	1	1	3	1	9	3
		11.7	27.1 + k	24.09		27.52 + 3.4k

Value of k = ?

$$\text{slope } b = 1.327$$

$$y = a + bx$$

$$\bar{y} = \frac{\sum y}{n} = \frac{27.1 + k}{8}$$

$$\bar{x} = \frac{11.7}{8} = 1.4625$$

$$b = \frac{SS_{xy}}{SS_{xx}}$$

$$SS_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n}$$

$$SS_{xy} = (27.52 + 3.4k) - \frac{(11.7)(27.1 + k)}{8}$$

$$SS_{xy} = \frac{236.16 + 27.2k - 317.07 - 11.7k}{8}$$

$$SS_{xy} = \frac{-80.91 - 15.5k}{8}$$

$$SS_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$SS_{xx} = 24.09 - \frac{(11.7)^2}{8}$$

$$SS_{xx} = 24.09 - 17.1125$$

$$SS_{xx} = 6.98$$

$$b = \frac{SS_{xy}}{SS_{xx}}$$

$$\text{Note } b = 1.327$$

$$1.327 = \frac{-80.91 - 15.5k}{6.98}$$

$$9.26 = -80.91 - 15.5k$$

$$94.1 = -80.91 - 15.5k$$

$$155.01 = -15.5k$$

$$k = \frac{155.01}{-15.5} = -10.000 \quad (C)$$

$$13. \quad y = a + bx$$

$$a = y - bx$$

$$a = \left(\frac{27.1 + k}{8} \right) - 1.327 (1.4625)$$

$$\text{recall } k = 10$$

$$a = \left(\frac{27.1 + 10}{8} \right) - 1.94$$

$$a = 4.64 - 1.94$$

$$a = 2.7 \quad (\text{A})$$

$$14. \quad y = a + bx$$

$$y = 3 + 1.3x \quad (\text{A})$$

$$15. \quad \bar{y} = \frac{\sum y}{n} = \frac{27.1 + k}{8} = \frac{27.1 + 10}{8}$$

$$\bar{y} = \frac{37.1}{8} = 4.64 \quad (\text{A})$$

$$16. \quad \frac{13 + 2y + 1 + (5y - 1) + 11}{5} = 9$$

$$\frac{13 + 2y + 1 + 5y - 1 + 11}{5} = 9$$

$$24 + 7y = 45$$

$$7y = 21$$

$$y = 3 \quad (\text{B})$$

$$17. \quad (\text{A})$$

$$18. \quad (\text{A})$$

$$19. \quad (\text{C})$$

20. $n = 148$, $\bar{x} = -7.1$, $\sigma = 1.2$

At 95% confidence interval

$$(1-\alpha) \% = (1 - 0.05) = 0.05$$

$$\frac{Z_{\alpha}}{2} = \frac{Z_{0.05}}{2} = Z_{0.025}$$

from Z table

$$Z_{0.025} = 1.960$$

$$\mu = \bar{x} \pm Z \frac{\sigma}{\sqrt{n}}$$

$$\mu = -7.1 \pm 1.960 \left(\frac{1.2}{\sqrt{148}} \right)$$

$$\mu = -7.1 \pm 1.960 (0.099)$$

$$\mu = -7.1 \pm 0.1933$$

$$\mu = -7.1 + 0.1933 \quad \text{OR} \quad \mu = -7.1 - 0.1933$$

$$\mu = -6.91 \quad \text{OR} \quad \mu = -7.2933$$

Ⓐ

21. $n = 16$, $\bar{x} = -7.1$, $S = 1.1$ At 95% Confidence interval

$$(1 - 0.05) = 0.05$$

$$\frac{Z_{0.05}}{2} = Z_{0.025} = 1.960$$

$$\mu = \bar{x} \pm Z \frac{\sigma}{\sqrt{n}}, \quad \mu = -7.1 \pm 1.960 \left(\frac{1.1}{\sqrt{16}} \right)$$

$$\mu = -7.1 \pm 1.960 (0.275)$$

$$\mu = -7.1 \pm 0.539$$

$$\mu = -7.1 + 0.539 \quad \text{OR} \quad \mu = -7.1 - 0.539$$

$$\mu = -6.561 \quad \text{OR} \quad \mu = -7.64$$

Ⓐ

22.

Staff Strength	No of Company	Cumulative frequency
1-10	1	1
11-20	5	6
21-30	10	16
31-40	19	35
41-50	42	77
51-60	10	87
61-70	6	93
71-80	4	97
81-90	2	99
91-100	1	100

$$\frac{N}{2} = \frac{100}{2} = 50$$

The class interval that falls immediate above $\frac{N}{2}$ is the median class

→ 41-50 is the median class

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f} \right) \times h$$

Where

L = lower limit of the median class

f = frequency of the median class

F = Cumulative frequency of the class preceding the median class

N = no of observation

h = width of the median class

$$\therefore \text{Median} = 41 + \left(\frac{\frac{100}{2} - 35}{42} \right) \times 9$$

$$= 44.21 \quad \text{(A)}$$

23.

Time	frequency
4.51 - 5.32	15
5.33 - 6.14	7
6.15 - 6.96	35
6.97 - 7.78	28
7.79 - 8.60	10
8.61 - 9.42	5

Modal class → 6.15 - 6.96

because it has the highest frequency

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

L = Lower limit, h = Size of interval

f₁ = frequency of modal class

f₀ = frequency of preceding class

f₂ = frequency of the class succeeding the modal class.

$$\text{Mode} = 6.15 + \left(\frac{35 - 7}{70 - 7 - 28} \right) \times 0.81$$

$$= 6.15 + \left(\frac{28}{35} \right) \times 0.81$$

$$= 6.798 \approx 6.8 \quad \text{(B)}$$

24.

x	0-9	10-19	20-29	30-39	40-49	50-59
f	10	2	4	5	25	14

Probability of the class interval 30-39 in the table?

Total sum of frequency = $10 + 2 + 4 + 5 + 25 + 14 = 60$

frequency of the class interval 30-39 is 5

$$\therefore \frac{5}{60} = \frac{1}{12} \quad (D)$$

25

x	f	x^2
25	7	625
29	9	841
$4+3$	8	3410.56
59	6	3481
70	20	4900
$186+y$	50	13257.56

$$\text{Mean } \bar{x} = 48.28$$

$$\bar{x} = \frac{\sum x}{n}$$

$$48.28 = \frac{186+y}{5}$$

$$241.4 = 186 + y$$

$$y = 55.4$$

$$S^2 = \frac{\sum x^2 - \left(\frac{\sum x}{n}\right)^2}{n-1} = \frac{13257.56 - \left(\frac{241.4}{50}\right)^2}{49}$$

$$S^2 = \frac{13257.56 - 23.31}{49} = \frac{13234.25}{49} = 270.1$$

$$\begin{aligned} \text{Standard deviation} &= \sqrt{\text{Variance}} \\ &= \sqrt{270.1} \\ &= 16.43 \end{aligned}$$

26.

x	f
$4+3$	8
70	20
25	7
59	6
29	9
$186+y$	50

$$\text{average } \bar{x} = 48.28$$

$$\bar{x} = \frac{\sum x}{n}, \quad 48.28 = \frac{186+y}{50}$$

$$2414 = 186 + y$$

$$y = 2228$$

27.

11-15	16-20	21-25	26-30	31-35	36-40
6	5	x	12	9	5

Sectorial angle of the class interval 26-30 ?

$$\frac{12}{100} \times 360 = 43.2$$

(A)

28.

x	26-30	31-35	36-40	41-45
frequency	p	7	10	21

Find the size of the distribution ?

The size of the distribution can be gotten from the class interval.

$$\rightarrow 30 - 26 = 4$$

$$35 - 31 = 4$$

$$40 - 36 = 4$$

$$45 - 41 = 4$$

$$\therefore 4 + 4 + 4 + 4 = 16$$

29. (C)

30.

x	frequency	mid point	mf	m ² f
15-19	3	17	51	867
20-24	4	22	88	1936
25-29	7	27	189	5103
30-34	5	32	160	5120
35-40	1	37.5	37.5	1406.25
	20		525.5	14432.25

$$\bar{x} = \frac{\sum mf}{n} = \frac{525.5}{20} = 26.28$$

(B)

31. Skewness = $3(\text{mean} - \text{median})$

class	mid point x	S.D		
		f	mf	m^2f
15-19	17	3	51	
20-24	22	4	88	867
25-29	27	7	189	1936
30-34	32	5	160	5103
35-40	37.5	1	37.5	5120
				1406.25

$$\text{Median} : \frac{N}{2} = \frac{20}{2} = 10$$

$\therefore$ Middle class $\rightarrow 25-29$

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f} \right) \times h = 25 + \left(\frac{\frac{20}{2} - 7}{7} \right) \times 4$$

$$\text{Median} = 25 + 1.714 = 26.71$$

$$s^2 = \frac{\sum m^2f}{N} - \left(\frac{\sum mf}{N} \right)^2$$

$$s^2 = \frac{14432.25}{20} - \left(\frac{525.5}{20} \right)^2 = \frac{13741.87}{2} = 687.09$$

Standard deviation = $\sqrt{\text{Variance}}$

$$S.D = \sqrt{687.09}$$

$$S.D = 26.21$$

$$\text{mean } \bar{x} = \frac{\sum mf}{N} = \frac{525.5}{20} = 26.28$$

$$\text{Skewness} = \frac{3(26.28 - 26.71)}{26.21}$$

$$= -0.049$$

32. x	F
0-5	2
5-15	6
15-25	20
25-35	15
35-45	8
45-55	4

L = lower limit, h = size of interval

f_1 = frequency of modal class

f_0 = frequency of the preceding class

f_2 = frequency of the class succeeding the modal class

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

modal class $\rightarrow 15-25$

$$\text{Mode} = 15 + \left(\frac{20 - 6}{40 - 6 - 15} \right) \times 10$$

$$\text{Mode} = 15 + 7.37 = 22.37 \quad (B)$$

33. Find the sum of the deviation from mean of the distribution

4, 6, 2, 8, 10, 3, 6, 4, 7, 10

STEP 1: Calculate the mean

$$\bar{x} = \frac{4+6+2+8+10+3+6+4+7+10}{10} = \frac{60}{10} = 6$$

STEP 2: Subtract the mean from each of the values

$$4-6 = -2$$

$$6-6 = 0$$

$$2-6 = -4$$

$$8-6 = 2$$

$$10-6 = 4$$

$$3-6 = -3$$

$$6-6 = 0$$

$$4-6 = -2$$

$$7-6 = 1$$

$$10-6 = 4$$

STEP 3: Square the difference between the values and the mean

$$(-2)^2 = 4$$

$$(4)^2 = 16$$

$$(1)^2 = 1$$

$$(0)^2 = 0$$

$$(-3)^2 = 9$$

$$(4)^2 = 16$$

$$(-4)^2 = 16$$

$$(0)^2 = 0$$

$$(2)^2 = 4$$

$$(-2)^2 = 4$$

Step 4: Sum up all the values of the square
 $4 + 16 + 4 + 16 + 9 + 4 + 1 + 16 = 70$

34. (A)

35. $d = 0.05$, $N = 28$, $\mu = 1.2$, $S = 2.14$
 $\bar{x} = 2$

$$Z = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}} = \frac{2 - 1.2}{\frac{2.14}{\sqrt{28}}} = \frac{0.8}{0.40} = 1.98 \quad (C)$$

36. (A)

37. (D)

38. Variance = 40

$$\frac{1}{n} \sum (\frac{1}{2} x_i - \frac{1}{2} \bar{x})^2$$

$$\frac{1}{n} \sum (\frac{1}{2})^2 (x_i - \bar{x})^2$$

$$\frac{1}{n} \sum \frac{1}{4} (x_i - \bar{x})^2$$

$$\frac{1}{4} \left(\frac{1}{n} \sum (x_i - \bar{x})^2 \right)$$

$$\frac{1}{4} (40) = 10 \quad (B)$$

39. $\text{Var}(x \pm n) = \text{Var}(x)$

$$\therefore \text{Var}(x - 2) = \text{Var}(x) = 30 \quad (B)$$

40. $n = 25$, $\bar{x} = 15.15$, $S = 1.5$

$$d = 95\% \quad (1 - 0.95) = 0.05, \quad Z \frac{0.05}{2} = 1.960$$

$$\mu = \bar{x} \pm Z \frac{S}{\sqrt{n}} = 15.15 \pm 1.960 \times \frac{1.5}{\sqrt{25}}$$

$$\mu = 15.15 \pm 0.588$$

$$\mu = 15.15 + 0.588 \quad \text{OR} \quad \mu = 15.15 - 0.588$$

$$\mu = 15.74 \quad \text{OR} \quad \mu = 14.56 \quad (A)$$

41. (B)

42. (B)

43. $\alpha = 7.5\%$

$$\alpha = \frac{7.5}{100} = 0.075$$

for two tailed $+Z_{\frac{\alpha}{2}} = \pm Z_{\frac{0.075}{2}}$

$$Z_{0.0375}$$

look for 0.0375 on the table or an approximate number

from Vertical $\rightarrow -0.08$

horizontal $\rightarrow \frac{-1.70}{-1.78}$ (C)

44. Normal distribution value = -1.72

horizontal $\rightarrow -1.70$

Vertical $\rightarrow \frac{-0.02}{-1.72}$

on the table it falls on 0.0427 (A)

45. $\bar{x} = 25.5$ min, $n = 25$, $\bar{x} = 25.5$

95% confidence interval

$$(1 - 0.95) = 0.05$$

$$Z_{\frac{\alpha}{2}} = Z_{\frac{0.05}{2}} = Z_{0.025} = 1.96$$

$$\mu = \bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} = 25.5 \pm 1.96 \left(\frac{2.5}{\sqrt{25}} \right)$$

$$\mu = 25.5 \pm 0.98$$

$$\mu = 26.48, \mu = 24.52 \quad (B)$$

46. 99% confidence interval

$$(1 - 0.99) = 0.01$$

$$Z_{\frac{\alpha}{2}} = Z_{\frac{0.01}{2}} = Z_{0.005} = 2.576$$

$$n = 36, \bar{x} = 15.5, s = 0.75$$

$$\mu = 15.5 \pm 2.576 \left(\frac{0.75}{\sqrt{36}} \right)$$

$$\mu = 15.5 \pm 0.322$$

$$\mu = 15.822 \text{ OR } \mu = 15.18 \quad (C)$$

47. At 95% Confidence interval

$$Z_{\frac{\alpha}{2}} = ? \quad (1 - 0.05) = Z_{0.05}$$

$$Z_{\frac{\alpha}{2}} = Z_{\frac{0.05}{2}} = Z_{0.025}$$

using t distribution

$$t_{0.025, n-1} = t_{0.025, 4}$$

$$= 2.776 \quad \textcircled{B}$$

48. Standard error of mean $Se = \frac{\sigma}{\sqrt{n}}$

from Question "49"

$$\text{Since } \bar{x} + 2 \frac{\sigma}{\sqrt{n}} = 5.27$$

$$10 + 2.776 \frac{\sigma}{\sqrt{n}} = 5.27$$

$$2.776 \frac{\sigma}{\sqrt{n}} = -4.73$$

$$\frac{\sigma}{\sqrt{n}} = -1.7 \quad \textcircled{C}$$

49. $\mu = 5.27$, $\mu = 14.73$, $n = 5$

At 95% Confidence interval

$$Z_{0.025, 4} = 2.776$$

$$\bar{x} + 2 \frac{\sigma}{\sqrt{n}} = 5.27 \quad \dots \textcircled{i}$$

$$\bar{x} - 2 \frac{\sigma}{\sqrt{n}} = 14.73 \quad \dots \textcircled{ii}$$

Adding equation $\textcircled{i}$ and $\textcircled{ii}$

$$2\bar{x} = 20$$

$$\bar{x} = 10 \quad \textcircled{D}$$

50.

$\textcircled{D}$

51. $\mu = 47.50, n = 25, s = 2.5, \bar{x} = 45$

1% confidence interval

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{45 - 47.50}{\frac{2.5}{\sqrt{25}}}$$

$$Z = \frac{-2.5}{0.5} = -5 \quad (\Delta)$$

52. $\alpha = 1\%$ confidence interval

$$Z_{\frac{\alpha}{2}} = Z_{\frac{0.01}{2}} = Z_{0.005} = 2.58 \quad (\Delta)$$

53. (C)

54. (B)

55. (A)

56. $x = 1, 3, m, 4, 5, 1, 1, 2, 3, 0$

$y = 5, 4, 2, 5, 1, 3, 4, 5, 5, 6$

$$\bar{y} = \frac{5+4+2+5+1+3+4+5+5+6}{10}$$

$$\bar{y} = \frac{40}{10} = 4$$

$$\therefore \bar{x} = 4$$

To find m

$$\bar{x} = \frac{\sum x}{n}, \quad 4 = \frac{20+m}{10}$$

$$40 = 20 + m$$

$$m = 20$$

$$s^2 \text{ of } x = (1-4)^2 + (3-4)^2 + (20-4)^2 + (4-4)^2 + (5-4)^2 + (1-4)^2 + (1-4)^2 + (2-4)^2 + (3-4)^2 + (0-4)^2$$

$$s^2 \text{ of } x = 9 + 1 + 256 + \frac{1+9+9+4+1+16}{9}$$

$$= 34 \quad (\text{C})$$

$$57. \text{Var}(x-y) = \text{Var}(x) + \text{Var}(y) - 2\text{Cov}(x, y)$$

$$\text{Var}(x) = 2\frac{1}{2} = \frac{5}{2}$$

$$\text{Var}(y) = 4$$

$$\text{Cov}(x, y) = \frac{1}{2}$$

$$\therefore \text{Var}(x-y) = \frac{5}{2} + 4 - 2\left(\frac{1}{2}\right)$$

$$= \frac{5}{2} + 4 - 1$$

$$= 5.5 \quad \textcircled{D}$$

$$58. \text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\text{but } \bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{1+3+3+5+5+6+8+9}{8} = 5$$

$$\text{and } \bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{2+3+5+4+6+5+7+9}{8} = 5$$

$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1-5 = -4	2-5 = -3	12
3-5 = -2	3-5 = -2	4
3-5 = -2	5-5 = 0	0
5-5 = 0	4-5 = -1	0
5-5 = 0	6-5 = 1	0
6-5 = 1	5-5 = 0	0
8-5 = 3	7-5 = 2	6
9-5 = 4	8-5 = 3	12
		<u>34</u>

$$\therefore \text{Cov}(x, y) = \frac{34}{8-1} = 4.86 \quad \textcircled{D}$$

$$59. \text{Coefficient of Variation (CV)} = \frac{\sigma}{\bar{x}} \times 100$$

$$\text{CV of computer students} = \frac{\sigma_y}{\bar{y}}$$

$$\sigma_y = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{-3^2 + (-2)^2 + \dots + 3^2}{8-1}}$$

$$\sigma_y = \sqrt{\frac{28}{7}} = 2$$

$$\therefore \text{Cl of Computer students} = \frac{2}{5} \times 100 = 40 \quad (\text{No answer})$$

60. (C)

$$61. n = 36, \bar{x} = 450.50, s = 100.50$$

$$\text{Standard error } SE = \frac{s}{\sqrt{n}} = \frac{100.50}{\sqrt{36}} = 16.75 \quad (D)$$

$$62. Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{400.00 - 450.50}{\frac{100.50}{\sqrt{36}}} = \frac{-50.5}{16.75}$$

$$Z = -3.01 \quad (B)$$

63. (C)

64. (A)

65. (B)

$$66. 30, 20, 16, 10, 24$$

$$S^2 = \frac{\sum (x - \bar{x})^2}{n-1}, \quad \bar{x} = \frac{100}{5} = 20$$

$$S^2 = \frac{(30-20)^2 + (20-20)^2 + (16-20)^2 + (10-20)^2 + (24-20)^2}{4}$$

$$S^2 = \frac{232}{4} = 58$$

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

$$SD = \sqrt{58} = 7.62 \quad (D)$$

$$67. \text{Standard error} = \frac{s}{\sqrt{n}} = \frac{7.62}{\sqrt{5}} = 3.41 \quad (C)$$

$$68. \bar{x} = 22, \mu = 20, \alpha = 5\%, \alpha = 0.05$$

$$Z_{\frac{0.05}{2}} = Z_{0.025} = 1.96$$

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{22 - 20}{\frac{7.62}{\sqrt{5}}} = 0.59 \quad (A)$$

69. (C)

70. (B)



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2018/2019

Name:..... Level: 100 Course Code: MTH 112 Title: Data Analysis
Duration: 2 hrs
Instruction: Answer all questions by shading the correct option in the question paper and copy the same option on the first page of the booklet. Submit both the question paper and the booklet for marking.

Use the following information to answer questions 1 to 3.
A random sample of 28 FUPRE students read a mean of 1.2 books with a standard deviation of 2.14 books in a month.

- TYPE A**
- At 0.05 level of significance, determine the calculated value of the statistic for testing the claim that the mean number of books read per month is less than 2
(a) -0.8 (b) 0.80 (c) 1.98 (d) -1.98
 - The table value for this test is
(a) 1.96 (b) 1.65 (c) 2.252 (d) 1.703
 - The decision for the hypothesis is
(a) Reserve judgment (b) Accept H_0 (c) Cannot reject H_0 (d) Reject H_0
 - The variance of a set of data is 40, what is the variance if each element of the data is divided by 2
(a) 40 (b) 10 (c) 160 (d) 80
 - The variance of a set of data is 30, what is the variance if 2 is subtracted from each element of the data
(a) 60 (b) 30 (c) 90 (d) 120

Use the following information to answer questions 6 to 8.
The mean price of a dozen roses at Shoprite from a random sample of size 30 is 48.33 with $S = 6.07$.

- The calculated value of the statistic for testing the claim that the mean price is higher than 45 at $\alpha = 1\%$
(a) 3.33 (b) 3.0 (c) 1.11 (d) 2.54
- What is the table value for this test
(a) 1.96 (b) 2.33 (c) 1.64 (d) 2.58
- What is the decision for the hypothesis
(a) Reserve judgment (b) Reject H_0 (c) Cannot reject H_0 (d) Accept H_0
- The volume of wine filled by a bottling machine is normally distributed with $\sigma = 5$ cl. In a sample of 16 bottles, $\bar{x} = 74.3$ cl was found. Construct a 95% confidence interval for μ
(a) (71.64, 76.96) (b) (71.85, 76.75) (c) (-71.64, 76.96) (d) (-71.85, 76.75)
- The 95% confidence interval for a data set with $n = 34$, $\bar{x} = 3.54$, $S = 0.13$ is
(a) (3.50, 3.58) (b) (-3.50, 3.58) (c) (3.40, 3.50) (d) (-3.40, 3.50)

Use the following information to answer questions 11 to 13.

If $\bar{x}_1 = 3$, $S_1 = 0.9$, $n_1 = 50$, $\bar{x}_2 = 3.5$, $S_2 = 1.2$, $n_2 = 50$.

- The value of the test statistic for testing the significant difference between the two sample means is -----
(a) 1.96 (b) 2.36 (c) -2.36 (d) -1.107
- What is the table value for testing the hypothesis using $\alpha = 1\%$
(a) 1.96 (b) 1.64 (c) 2.58 (d) 2.24
- What is the decision for the hypothesis using $\alpha = 1\%$
(a) Reserve judgment (b) Cannot accept H_0 (c) Reject H_0 (d) Accept H_0
- A company claims that its meat pie takes 10 minutes to make. To test if the actual time taken is more than the company claim, samples were selected and hypothesis testing carried out. What type of test is this
(a) type I (b) type II (c) two-tailed (d) one-tailed
- H_0 for testing hypothesis using normal distribution is rejected if
(a) $P \text{ value} \leq \alpha$ (b) $P \text{ value} \geq \alpha$ (c) $P \text{ value} > \alpha$ (d) $P \text{ value} < \alpha$

Use the following information to answer questions 16 to 18.

A1 OF A4

- A random sample of 34 Shoprite customers spent a mean amount of 4.43 and standard deviation of 3.23.
16. The calculated value of the statistic for testing the claim that the mean amount spent is less than 5 at 0.05 level of significance is (a) 1.03 (b) 1.11 (c) -1.03 (d) -1.11
17. What is the table value for this test (a) 1.96 (b) 2.33 (c) 1.65 (d) 2.58
18. What is the decision for the hypothesis (a) Reserve judgment (b) Reject H_0 (c) Accept H_0 (d) Cannot accept H_0
19. Acceptance of the null hypothesis when it is false leads to ----- error
(a) type I (b) type II (c) one-tailed (d) two-tailed
20. The first step in hypothesis testing is the determination of -----
(a) critical region (b) H_0 and H_1 (c) calculated value of the test statistic (d) decision
- Use the information on colour preference by gender in the table to answer questions 21 to 25.

Gender	Colour			
	Bright	Light	Dull	Other
Male	15	18	35	15
Female	35	12	10	10

21. How many males should we have expected to prefer light colour?
(a) 16.6 (b) 27.67 (c) 1.66 (d) 13.83
22. How many females should we have expected to prefer bright colour?
(a) 22.33 (b) 27.67 (c) 16.6 (d) 13.40
23. What is the calculated value for testing the dependent of colour on gender
(a) 24.007 (b) 27.672 (c) 16.642 (d) 23.405
24. What is the table value for this test
(a) 7.815 (b) 2.58 (c) 2.131 (d) 1.96
25. What is the decision for the hypothesis
(a) Reject H_0 (b) Accept H_0 (c) Cannot reject H_0 (d) Reserve judgment
26. If $n = 148$, $\bar{x} = -7.1$, $\sigma = 1.2$. A 95% confidence interval for μ is -----
(a) (6.91, 7.29) (b) (-7.29, -6.91) (c) (-7.29, 6.91) (d) (-6.91, -7.29)
27. If $n = 16$, $\bar{x} = -7.1$, $S = 1.1$. A 95% confidence interval for μ is -----
(a) (6.51, 7.69) (b) (-7.69, -6.51) (c) (-6.51, 7.69) (d) (6.69, 7.51)
- Use the following information to answer questions 28 to 29.
If $\bar{x}_1 = 8.5$, $S_1 = 2.3$, $n_1 = 100$, $\bar{x}_2 = 7.5$, $S_2 = 2.8$, $n_2 = 110$.
28. The value of the test statistic for testing the significant difference between the two sample means is -----
(a) 1.96 (b) -2.84 (c) 2.84 (d) 2.58
29. What is the decision for the hypothesis using $\alpha = 5\%$
(a) Accept H_0 (b) Cannot reject H_0 (c) Reject H_0 (d) Reserve judgment
30. If $\text{Var}(X) = 2$, then $\text{Var}(X + C)$ where $C = 2$ is ----- (a) 6 (b) 4 (c) 8 (d) 2
- Use the following information to answer questions 31 to 33.
A 95% confidence interval reported for the parameter μ is 1.6 to 7.8. The interval is based on 16 observations constructed using t-distribution.
31. What is the table value of the t-distribution
(a) 2.131 (b) 1.64 (c) 2.58 (d) 2.24
32. What is the value of the standard error of mean ($s.e(\bar{x})$)
(a) 1.45 (b) 2.131 (c) 4.18 (d) 1.64
33. What is the mean of the unknown data set
(a) 4.70 (b) 1.60 (c) 1.50 (d) 2.54
34. If $\alpha = 2.5\%$, what is the table value for a two-tailed test using normal distribution
(a) 1.96 (b) 1.64 (c) 2.58 (d) 2.24
35. The critical region is the region for
(a) Acceptance of H_0 (b) Acceptance of H_1 (c) Rejection of H_1 (d) Rejection of H_0
36. The use of sample for taking decision about the population is refer to as ----- statistic.
(a) Descriptive (b) Estimation (c) Sampling (d) Inferential
37. Tabular, graphical and numerical methods used to summarize data set is called ----- statistics
(a) Data presentation (b) Inferential (c) Descriptive (d) bar chart

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38. In measuring sample, we normally summarize the sample ----- to be able to generalize the population.
 (a) Variable (b) parameter (c) statistic (d) survey
39. When equal interval is used in measuring population or sample, it is better to used -----
 (a) ordinal scale (b) ratio scale (c) nominal scale (d) Interval scale
40. If $\text{Var}(X) = 4$, then $\text{Var}(X - C)$ where $C = 2$ is ----- (a) 6 (b) 4 (c) 2 (d) 4
41. Calculate x if the mean of 30, $(x + 5)$, 12, 40, and 10 is equal to $(x + 1)$
 (a) 25 (b) 25.5 (c) 23 (d) 22
42. ----- is a good example of sampling frame. (A) Voter's card (B) Voter's register (C) Class attendance
 (D) Employee's name
43. Why is sample preferred to population in most statistical enquiry? (A) Biased (B) Statistic
 (C) Unbiased (D) Parameter
44. Select a sample of 5 members from a population consisting of 20 members. Given that the first number select is 3. Find the mean of the sample using systematic random sampling (A) 11 (B) 3 (C) 7 (D) 19
45. Which of the following is a quasi-homogeneous random sampling? (A) Systematic random sampling (B) Simple random sampling (C) Stratified random sampling (D) Cluster sampling
46. ----- is a sampling method drawn from different groups from a given population. (A) Multi-stage sampling (B) Stratified random sampling (C) Cluster sampling (D) Systematic random sampling
47. A survey was carried out on some students of the MTH 112 to assess their performance. What is this type of sampling called? (A) Simple random sampling (B) Cluster sampling
 (C) Multi-stage sampling (D) Pilot survey
48. A measurement scale that serve as a label is term (A) Ordinal (B) Ratio (C) Interval
 (D) Nominal
- Given the follow sample distribution table
- | | | | | |
|----------------|-------|-------|-------|-------|
| Class interval | 31-40 | 41-50 | 51-60 | 61-70 |
| Frequency | 6 | 10 | 3 | 1 |
49. What is the upper class boundary of the second class (A) 40.5 (B) 50.5 (C) 60.5 (D) 30.5
50. The following are marks scored by some students in 100 level
 14, 19, 18, 12, 18, 8, 19, 16, 10, 11, 3, 17, 18, 11, 15, 5, 20, 16, 12, 3.
 Find the frequency mark for a class interval 16-20, using the class size of 5. (A) 4 (B) 2 (C) 9 (D) 8
51. Which of the following is a common factor to bar chart and histogram (A) Class interval
 (B) Class boundary (C) Frequency (D) Class marks
52. A chemist has undertaken an experiment where they tried adding 6 different amounts of a catalyst in a mixture is given as 4.1, 2.0, 6.0, 2.8, 1.1, and 4.9. Find $\sum_{i=1}^n (x_i - \bar{x})$. (A) 20.9 (B) 3.0 (C) 6.97 (D) 0
53. Which of the following is the variance of y ? (A) $\sum_{i=1}^n y_i^2 - \bar{y}$ (B) $\sum_{i=1}^n y_i^2 - \sum_{i=1}^n y_i$
 (C) $\sum_{i=1}^n (y_i - \bar{y}) y_i$ (D) $\sum_{i=1}^n y_i^2 - \bar{y}^2$
54. Given the following data 60, 49, 40, 61, 64, 60, 59, 54, 62, 69, 70, 42, 56, 61, 61, 61, 58, 51, 48, 65, 49, 49, 41, 48, 52, 46, 59, 46, 58, 43. What is the sector angle of 69? (A) 4 (B) 10.8
 (C) 14.4 (D) 7.2
55. Find the semi-interquartile range of 10 students scores in PHY 109 given below 10, 6, 6, 2, 10, 3, 7, 4, 8, 4 (A) 3.75 (B) 8.5 (C) 4.75 (D) 2.38
56. Which of the following best explain why we classify data? (A) to prepare tabulation (B) to eliminate necessary details (C) to draw inference (D) to present data in minimum space
57. What is the mode of the following numbers 1, 8, 8, 10, 9, 2, 7, 8, 2, 2, 4, 1, 1, 8, 7, 1 (A) 8 (B) 2 (C) 1 (D) None of the above
58. The standard deviation of a statistic describes (A) the shape of its distribution (B) the centre of its distribution (C) the amount of skewness associated with its distribution (D) the amount of variability associated with its distribution
59. One can easily determine the ... of a distribution from histogram (A) mean (B) mode (C) median (D) standard deviation

A3 OF A4

60. An experiment in which the outcomes are two probabilities: Success or Failure is said to be. (A) Binomial (B) Normal (C) Geometric (D) Bernoulli
61. The formula for spearman's rank correlation is (A) $1 + \frac{6 \sum d^2}{n(n^2-1)}$ (B) $1 - \frac{\sum d^2}{n(n^2-1)}$ (C) $1 - \frac{6 \sum d^2}{n(n^2-1)}$ (D) $1 - \frac{6 \sum d^2}{n^2}$
62. A number is selected at random from 0 to 20. What is the probability that the number is an odd prime? (A) $\frac{2}{5}$ (B) $\frac{7}{20}$ (C) $\frac{1}{2}$ (D) $\frac{1}{20}$
63. The range of values a probability can assume is: (A) from 0 to 1 (B) from -1 to 1 (C) from 1 to 100 (D) from 0 to $\frac{1}{2}$
64. Let P be the set of prime factors 42 and Q be the set of prime factors of 45. Find $P \cap Q$ (A) {2} (B) {3} (C) {7} (D) {5}
65. A survey was carried out on some students of the MAT 004 to assess their performance. What is this type of sampling called? (A) Simple random sampling (B) Cluster sampling (C) Multi-stage sampling (D) Pilot survey
66. A random distribution that measures rare events in space and time that has its mean and variance the same is expressed as (A) Poisson (B) Beta (C) Gamma (D) Hypergeometric
67. Which of the following is a common factor to bar chart and histogram (A) Class interval (B) Class boundary (C) Frequency (D) Class marks
68. The following are marks scored by some students in 100 level
14, 19, 18, 12, 18, 8, 19, 16, 10, 11, 3, 17, 18, 11, 15, 5, 20, 16, 12, 3.
Find the frequency mark for a class interval 16-20, using the class size of 5. (A) 4 (B) 2 (C) 9 (D) 8
Given the follow sample distribution table
- | | | | | |
|----------------|-------|-------|-------|-------|
| Class interval | 31-40 | 41-50 | 51-60 | 61-70 |
| Frequency | 6 | 10 | 3 | 1 |
69. What is the upper class boundary of the second class (A) 40.5 (B) 50.5 (C) 60.5 (D) 30.5
70. ----- is a good example of sampling frame. (A) Voter's card (B) Voter's register (C) Class attendance (D) Employee's name

Best of Luck.

Examiners: S. O. Ogundele, J. T. Eghwerido.

2018/2019 EXAM (TYPE A)

1) $n = 28, \bar{x} = 1.2, s = 2.14, \alpha = 0.05, \mu_0 = 2, H_0: \mu < \mu_0$
 since $n < 30$, we use "t" distribution

$$t_{cal} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1.2 - 2}{2.14/\sqrt{28}} = -1.98 \text{ (D)}$$

2) $t_{Tab} = t_{(1-\alpha), (n-1)} \rightarrow$ for one tailed test

$\Rightarrow t_{(1-0.05), (28-1)} = t_{(0.95), (27)} = 1.703 \text{ (D)}$

Note: $t_{(0.95)} \Leftrightarrow t_{(0.05)}$

3) Since $t_{cal} < t_{Tab}$ we accept H_0 (B)

4) $Var(n\bar{x}) = n^2 \times Var(x)$

$\therefore Var(\frac{x}{4}) = \frac{1}{4} \times Var(x) \Rightarrow$ where $x = 40$

$\therefore \frac{1}{4} \times 40 = 10 \text{ (B)}$

5) $Var(x \pm n) = Var(x) \therefore Var(x-2) = Var(x) = 30 \text{ (B)}$

6) $n = 30, \bar{x} = 48.33, s(\sigma) = 6.07, H_0: \mu > \mu_0$

$H_1: \mu \neq \mu_0, \mu_0 = 45, \alpha = 1\% = 0.01$

Since $n = 30$ and " σ " is known, we use "Z" distribution

$$Z_{cal} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{48.33 - 45}{6.07/\sqrt{30}} = 3.005 \text{ (B)}$$

7) $Z_{Tab} = Z_{\alpha}$ for one-tailed test

$\therefore Z_{(0.01)} = 2.33 \text{ (B)}$

8) Since $Z_{cal} > Z_{Tab}$, that is $\mu > \mu_0$, we Accept H_0 (D)

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(1)

9) $\sigma = 5$, $n = 16$, $\bar{x} = 74.3$, $\alpha = 95\% = 0.95 = 1 - 0.05 = 0.95$
 Confidence interval for $\mu = \bar{x} \pm t_{(1-\frac{\alpha}{2}), (n-1)} \times \left(\frac{\sigma}{\sqrt{n}}\right)$

$$\mu = 74.3 \pm t_{(1-\frac{0.05}{2}), (16-1)} \times \frac{5}{\sqrt{16}} \Rightarrow 74.3 \pm t_{(0.975), (15)} \times \left(\frac{5}{4}\right)$$

$$\mu = 74.3 \pm (2.131 \times \frac{5}{4}) \Rightarrow (71.64, 76.96) \text{ (A)}$$

10) $\alpha = 95\% = 0.95$, $n = 34$, $\bar{x} = 3.54$, $s(\sigma) = 0.13$
 $1 - 0.05 = 0.95$

$$\mu = 3.54 \pm Z_{(1-\frac{0.05}{2})} \times \frac{0.13}{\sqrt{34}} \Rightarrow 3.54 \pm Z_{(0.975)} \times (0.0223)$$

$$\mu = 3.54 \pm 1.96 \times 0.0223 = (3.50, 3.58) \text{ (A)}$$

11) $\bar{x}_1 = 3$, $s_1 = 0.9$, $n_1 = 50$, $\bar{x}_2 = 3.5$, $s_2 = 1.2$, $n_2 = 50$

$$Z_{cal} = \frac{\bar{x}_1 - \bar{x}_2 - C}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad \rightarrow C = 0 \text{ if not stated or given}$$

$$= \frac{3 - 3.5 - 0}{\sqrt{\frac{0.9^2}{50} + \frac{1.2^2}{50}}} = \frac{-0.5}{0.2121}$$

$$Z_{cal} = -2.357 \text{ (C)}$$

12) at $\alpha = 1\% = 0.01$, $Z_{Tab} = Z_{(1-\frac{\alpha}{2})}$

$$Z_{Tab} = Z_{(1-\frac{0.01}{2})} = Z_{(0.995)} = 2.58 \text{ (C)}$$

13) Since $(Z_{cal} = -2.357)$ falls in between $(Z_{Tab} = -2.58 \leftrightarrow +2.58)$
 we Accept H_0 (D)

Note: we used Z_{Tab} i.e. from $-2.58 \rightarrow +2.58$
 because it's a two-tailed Test



14) since we are testing if the time is more (ie greater than) it is a one-tailed Test (D)

15) H_0 is rejected if P-value $< \alpha$ (D)

16) $n=34$, $\bar{x}=4.43$, $\sigma=3.23$, $\mu_0=5$, $\alpha=0.05$, $H_0: \mu < \mu_0$

$$Z_{cal} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{4.43 - 5}{3.23/\sqrt{34}} = -1.03 \text{ (C)}$$

17) $Z_{tab} = Z_{(1-\alpha)}$ for one-tailed Test

$$Z_{tab} = Z_{(1-0.05)} = Z_{(0.95)} = 1.65 \text{ (C)}$$

18) since $Z_{cal} < Z_{tab}$, Accept H_0 (C)

19) (B)

20) (B)

Questions 21 to 25

Gender	colour				Total
	Bright	Light	Dull	others	
Male	15	18	35	15	83
Female	35	12	10	10	67
Total	50	30	45	25	150

Calculating the expected frequency

$E_{gc} = \frac{G_i C_j}{G.T}$ where $E_{gc} \rightarrow$ Expected frequency

$G_i \rightarrow$ Total no of male or female

$C_j \rightarrow$ Total no in colour column

$G.T \rightarrow$ sum of Rows and column = 150

$$E_{MB} = \frac{G_{mC_B}}{G.T} = \frac{83 \times 50}{150} = 27.67$$

$\downarrow$ male Bright

$$E_{ML} = \frac{G_m C_L}{G_T} = \frac{83 \times 30}{1150} = 16.6; E_{MD} = \frac{G_m C_D}{G_T} = \frac{83 \times 45}{150} = 24.9$$

$$E_{MO} = \frac{83 \times 25}{150} = 13.83$$

$$E_{FB} = \frac{G_F C_B}{G_T} = \frac{67 \times 50}{150} = 22.33$$

Female Bright

$$E_{FL} = \frac{67 \times 30}{150} = 13.4; E_{FD} = \frac{67 \times 45}{150} = 20.1$$

$$E_{FO} = \frac{67 \times 25}{150} = 11.17$$

O	E	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
15	27.67	-12.67	160.5289	5.8016
18	16.6	1.4	1.96	0.1181
35	24.9	10.1	102.01	4.0968
15	13.83	1.17	1.3689	0.0990
35	22.33	12.67	160.5289	7.1890
12	13.4	-1.4	1.96	0.1463
10	20.1	-10.1	102.01	5.0751
10	11.17	-1.17	1.3689	0.1226

22.6485

From the above calculations;

21) No. of male expected to prefer light colour = $E_{ML} = 16.6$ (A)

22) $E_{FB} = 22.33$ (A)

23) The calculated value is given by $\chi^2 = \sum \frac{(O-E)^2}{E}$

$$\Rightarrow 22.64 \approx 23 \text{ (D)}$$

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24) Table value is given by $\chi^2_{(r-1)(c-1), \alpha}$
 where $r \rightarrow$ no of rows = 2
 $c \rightarrow$ no of columns = 4, $\alpha = 0.05$ {Assumed}
 $\therefore \chi^2_{(2-1)(4-1), (0.05)} = \chi^2_{(3), (0.05)} = \underline{7.815}$ (A)

25) Since $X^2_{cal} > X^2_{Tab}$, we reject H_0 (A)

26) $n=148$, $\bar{x} = -7.1$, $\sigma = 1.2$, $\alpha = 95\% = 0.95$, $1-0.95 = 0.05$
 $\mu = \bar{x} \pm Z_{(1-\frac{0.05}{2})} \times \frac{\sigma}{\sqrt{n}}$

$$\mu = -7.1 \pm (1.96) \times \frac{1.2}{\sqrt{148}} \Rightarrow -7.1 \pm (0.1933)$$

$$\Rightarrow \underline{-7.27, -6.91}$$
 (D)

27) $n=16$, $\bar{x} = -7.1$, $s = 1.1$, $\alpha = 95\% = 0.95$, $1-0.95 = 0.05$

$$\mu = \bar{x} \pm t_{(1-\frac{\alpha}{2}), (n-1)} \times \frac{s}{\sqrt{n}}$$

$$\mu = -7.1 \pm t_{0.975, 15} \times \frac{1.1}{\sqrt{16}} \Rightarrow -7.1 \pm (2.131 \times 0.275)$$

$$\Rightarrow \underline{(-7.69, -6.51)}$$
 (B)

28) $\bar{x}_1 = 8.5$, $s_1 = 2.3$, $n_1 = 100$, $\bar{x}_2 = 7.5$, $s_2 = 2.8$, $n_2 = 110$

$$Z_{cal} = \frac{\bar{x}_1 - \bar{x}_2 - c}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{8.5 - 7.5 - 0}{\sqrt{\frac{(2.3)^2}{100} + \frac{(2.8)^2}{110}}} = \underline{2.838}$$
 (C)

29) at $\alpha = 5\% = 0.05$, $Z_{Tab} = Z_{(1-\frac{\alpha}{2})} \Rightarrow Z_{(1-\frac{0.05}{2})}$

$$\Rightarrow Z_{0.975} = 1.96$$

Since $Z_{cal} > Z_{Tab}$, we reject H_0 (C)

30) $\text{Var}(X+n) = \text{Var}(X)$ where $n = \text{constant}$
 $\therefore \text{Var}(X+2) = \text{Var}(X) = 2$ (D)

31) $\alpha = 95\%$, $\mu = 1.6 \rightarrow 7.8$, $n = 16$
 $1 - 0.95 = 0.05$

$\therefore t_{(1-\frac{\alpha}{2}), (n-1)} = t_{(1-\frac{0.05}{2}), (16-1)} = t_{(0.975), 15} = \underline{2.131}$ (A)

32) To find S.E($\bar{x}$)

we solve simultaneously

$\mu = \bar{x} \pm t_{\text{tab}} \times \text{S.E}(\bar{x})$ Please note $\text{S.E}(\bar{x}) = \frac{s}{\sqrt{n}}$

$\therefore 7.8 = \bar{x} + 2.131 \text{ S.E}(\bar{x})$ — (1)

$1.6 = \bar{x} - 2.131 \text{ S.E}(\bar{x})$ — (2)

by subtracting eqn (2) from eqn (1), we have:

$6.2 = 4.262 \text{ S.E}(\bar{x})$

$\therefore \text{S.E}(\bar{x}) = \frac{6.2}{4.262} = \underline{1.45}$ (A)

33) To find mean ($\bar{x}$)

substitute the value of S.E($\bar{x}$) into eqn (1)

i.e. $7.8 = \bar{x} + 2.131(1.45)$

$\therefore \bar{x} = 7.8 - 3.09 = \underline{4.71}$ (A)

34) if $\alpha = 2.5\% = 0.025$, using normal "Z" distribution

$Z_{\text{tab}} = Z_{(1-\frac{\alpha}{2})} = Z_{(1-\frac{0.025}{2})} = Z_{(0.9875)} =$

35) (D)

36) (D)

37) (C)

38) (C)

39) (D)

40) $\text{Var}(x-n) = \text{Var}(x) = \underline{4}$ (B)



$$41) \bar{x} = \frac{\sum x}{n} \Rightarrow \frac{30 + (x+5) + 12 + 40 + 10}{5} = x+1$$

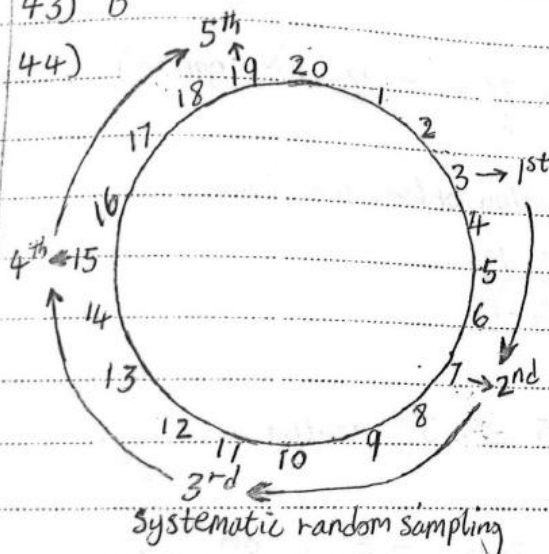
where $\bar{x} = x+1$ and $n = 5$

$$\therefore \frac{97+x}{5} = x+1 \Rightarrow 97+x = 5x+5 \Rightarrow x = \underline{23} \text{ (C)}$$

42) (B)

43) B

44)



$$N = 20, n = 5$$

$$k = N/n = \frac{20}{5} = 4$$

$$1^{\text{st}} \text{ member} = 3$$

$$2^{\text{nd}} \text{ member} = 3 + k = 3 + 4 = 7$$

$$3^{\text{rd}} = 3 + k + k = 11$$

$$4^{\text{th}} = 3 + k + k + k = 15$$

$$5^{\text{th}} = 3 + 4k = 19$$

$$\therefore \bar{x} = \frac{\sum x}{n} = \frac{3 + 7 + 11 + 15 + 19}{5} = \underline{11} \text{ (A)}$$

45) D

46) B

47) A

48) D

49) B

50) C

51) C

$$52) \bar{x} = \frac{4.1 + 2.0 + 6.0 + 2.8 + 1.1 + 4.9}{6} = 3.48 \approx 3.5$$

$$\begin{aligned} \therefore \sum_{i=1}^n (x_i - \bar{x}) &= (4.1 - 3.5) + (2.0 - 3.5) + (6.0 - 3.5) + (2.8 - 3.5) \\ &\quad + (1.1 - 3.5) + (4.9 - 3.5) \\ &= \underline{\underline{0.02}} \text{ (D)} \end{aligned}$$

53) A

$$54) \text{ sector angle of } 69 = \frac{1}{30} \times 360 = \underline{\underline{12}} \text{ (No option)}$$

55) 1st arrange the scores in ascending order, we have;

$$\begin{array}{cccccccccc} 2, & 3, & 4, & 4, & 6, & 6, & 7, & 8, & 10, & 10 \\ 1^{st} & 2^{nd} & 3^{rd} & 4^{th} & 5^{th} & 6^{th} & 7^{th} & 8^{th} & 9^{th} & 10^{th} \end{array}$$

$$\text{Semi-Interquartile Range} = \frac{Q_3 - Q_1}{2}$$

To find Q_1 ;

$$\text{we use } \frac{N}{4} = \frac{10}{4} = 2.5 \approx 3^{rd} \text{ position}$$

$$\therefore Q_1 = 4$$

$$\text{To find } Q_3; \text{ we use } \frac{3N}{4} = \frac{3 \times 10}{4} = 7.5 \approx 8^{th} \text{ position}$$

$$\therefore Q_3 = 8$$

$$\rightarrow \text{The semi-Interquartile Range} = \frac{Q_3 - Q_1}{2} = \frac{8 - 4}{2} = \underline{\underline{2}} \text{ (D)}$$

56) (C)

57) (C)

58) (D)

59) (B)

60) A

61) (C)

62) The odd prime numbers between 0 and 20 are
3, 5, 7, 11, 13, 17 and 19 i.e. 7 in total
 $\therefore$ The probability is $\frac{7}{20}$ (B)

63) (A)

64) set "P" Prime factors of 42 = $\{2, 3, 7\}$
set Q Prime factors of 45 = $\{3, 5\}$
 $\therefore P \cap Q = \{3\}$ (B)

65) (A)

66) (A)

67) (C)

68) (C)

69) (B)

70) (C)



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2017/2018

Name: Level: 100 Course Code: MTH 112 Title: Data Analysis
Duration: 2 hrs
Instruction: Answer all questions by shading the correct option in the question paper and copy the same option on the first page of the booklet. Submit both the question paper and the booklet for marking.

Use the following distribution below for questions 1 to 3.
TYPE B

Number of Hours	0 - 9	10 - 19	20 - 29	30 - 39	40 - 49	50 - 59
Frequency	40	50	70	40	30	20

- What is the mode of the above distribution? (A) 17.5 (B) 29.5 (C) 26.5 (D) 23.5
- What is the first quartile of the above distribution? (A) 12 (B) 14 (C) 16 (D) 18
- What is the median of the above distribution? (A) 20.5 (B) 23.5 (C) 24.5 (D) 25.5
- The arithmetic mean of a set of 5 numbers is 20. If each number is first multiplied by 2 and then added by 5, then what is the mean of new numbers? (A) 11.5 (B) 22.5 (C) 33.5 (D) 44.5
- What is the weighted mean of first 10 natural numbers whose weights are equal to the corresponding number? (A) 7 (B) 5.5 (C) 5 (D) 4.5
- The sum of mean, mode and median of the data 12, 15, 11, 13, 18, 11, 13, 12, 13 to the nearest whole number is (A) 39 (B) 36 (C) 37 (D) 38
- The arithmetic mean of the first ten whole numbers is (A) 5.5 (B) 5 (C) 4 (D) 4.5
- If the arithmetic mean of the following data $6 - x, 7 + 2x, 3 - 3x, 5 + 4x, 4 - 5x, 11 + 6x, 2 - 7x, 18 + 8x$ is 10, what is the value of x ? (A) 5 (B) 6 (C) 7 (D) 4
- What is the geometric mean of the following data 4, 7, 3, 5, 3, 6, 4, 4, 5, 4, 6, 6, 8? (A) 4.33 (B) 4.79 (C) 4.46 (D) 4.70

Use the following distribution below for questions 10-15

Class Interval	13 - 17	18 - 22	23 - 27	28 - 32	33 - 37	38 - 42	43 - 47	48 - 52	53 - 57
Frequency	2	22	10	14	3	4	6	1	1

- Find the standard deviation of the above distribution (A) 19.40 (B) 16.30 (C) 9.40 (D) 11.39
- What is the variance? (A) 88.45 (B) 93.60 (C) 77.40 (D) 69.70
- What is the coefficient of variation? (A) 33.27 (B) 26.14 (C) 39.70 (D) 63.40
- Determine the degree of asymmetry of the distribution (A) 0.0617 (B) 0.0120 (C) -0.0617 (D) -0.0566
- What is the cumulative frequency of 43 - 47 class interval? (A) 6 (B) 57 (C) 61 (D) 63
- What is the cumulative frequency before the third quartile class? (A) 34 (B) 48 (C) 24 (D) 51
- What is the harmonic mean of the following data 4, 7, 3, 5, 3, 6, 4, 4, 5, 4, 6, 6, 8? (A) 4.23 (B) 4.58 (C) 4.74 (D) 4.30
- Suppose that the mean and median of a given distribution are 63.7 and 62.5 respectively, determine the mode. (A) 60.1 (B) 63.7 (C) 62.5 (D) 61.2
- Find the mean deviation of the following data 3, 5, 6, 7, 8, 6, 10, 12 (A) 2.12 (B) 2.16 (C) 2.30 (D) 2.40
- The sum of the percent frequencies for all classes will always equal to (A) one (B) the number of classes (C) the number of items in the study (D) 100
- Since the mode is the most frequency occurring data value, then (A) it can never be larger than the mean (B) it is always greater than the median (D) none of the above
- If the semi-interquartile range value and third quartile values are 12.9 and 66 respectively, find the value of the first quartile (A) 30.2 (B) 40.2 (C) 20.3 (D) 35.6

22. A series of numerical figures which show the relative position is called (A) Index number (B) Relative number (C) Absolute number (D) None of the above
23. Index number for the base period is always taken as (A) 200 (B) 50 (C) 1 (D) 100
24. Index numbers are often constructed from the (A) frequency (B) Class (C) sample (D) None
25. The ratio of price of single commodity in a given period to its price in another period is called the (A) Base period (B) Price ratio (C) Relative price (D) None
26. ----- is a good example of sampling frame. (A) Voter's card (B) Voter's register (C) Class attendance (D) Employee's name
27. Why is sample preferred to population in most statistical enquiry? (A) Biased (B) Statistic (C) Unbiased (D) Parameter
28. Select a sample of 5 members from a population consisting of 20 members. Given that the first number select is 3. Find the mean of the sample using systematic random sampling (A) 11 (B) 3 (C) 7 (D) 19
29. Which of the following is a quasi-homogeneous random sampling? (A) Systematic random sampling (B) Simple random sampling (C) Stratified random sampling (D) Cluster sampling
30. ----- is a sampling method drawn from different groups from a given population. (A) Multi-stage sampling (B) Stratified random sampling (C) Cluster sampling (D) Systematic random sampling
31. A survey was carried out on some students of the MTH 112 to assess their performance. What is this type of sampling called? (A) Simple random sampling (B) Cluster sampling (C) Multi-stage sampling (D) Pilot survey
32. A measurement scale that serve as a label is term (A) Ordinal (B) Ratio (C) Interval (D) Nominal
- Given the following sample distribution table
- | | | | | |
|----------------|-------|-------|-------|-------|
| Class interval | 31-40 | 41-50 | 51-60 | 61-70 |
| Frequency | 6 | 10 | 3 | 1 |
33. What is the upper class boundary of the second class (A) 40.5 (B) 50.5 (C) 60.5 (D) 30.5
34. The following are marks scored by some students in 100 level
14, 19, 18, 12, 18, 8, 19, 16, 10, 11, 3, 17, 18, 11, 15, 5, 20, 16, 12, 3.
Find the frequency mark for the class interval 16-20, using the class size of 5. (A) 4 (B) 2 (C) 9 (D) 8
35. Which of the following is a common factor to bar chart and histogram (A) Class interval (B) Class boundary (C) Frequency (D) Class marks
36. A chemist has undertaken an experiment where he tried adding 6 different amounts of a catalyst in a mixture given as 4.1, 2.0, 6.0, 2.8, 1.1, and 4.9. Find $\sum_{i=1}^n (x_i - \bar{x})$. (A) 20.9 (B) 3.0 (C) 6.97 (D) 0
37. Which of the following is the variance of y? (A) $\sum_{i=1}^n y_i^2 - \bar{y}$ (B) $\sum_{i=1}^n y_i^2 - \sum_{i=1}^n y_i$ (C) $\sum_{i=1}^n (y_i - \bar{y}) y_i$ (D) $\sum_{i=1}^n y_i^2 - \bar{y}^2$
38. Given the following data 60, 49, 40, 61, 64, 60, 59, 54, 62, 69, 70, 42, 56, 61, 61, 61, 58, 51, 48, 65, 49, 41, 48, 52, 46, 59, 46, 58, 43. What is the sector angle of 69? (A) 4 (B) 10.8 (C) 14.4 (D) 7.2
39. Find the semi-interquartile range of 10 students scores in PHY 109 given below 10, 6, 6, 2, 10, 3, 7, 8, 4 (A) 3.75 (B) 8.5 (C) 4.75 (D) 2.38
40. Which of the following best explain why we classify data? (A) to prepare tabulation (B) to eliminate necessary details (C) to draw inference (D) to present data in minimum space
41. If $\alpha = 1.5\%$, what is the table value for a two-tailed large sample test.
(A) 1.96 (B) 2.43 (C) 2.17 (D) 1.645
42. A test of the degree of association based on ranking is carried out using (A) regression (B) Spearman's method (C) Pearson's method (D) product moment.
43. Given the fitted regression model $y = 20 - 4x$. The relationship between x and y is (A) positive (B) direct (C) inverse (D) equal
44. If the correlation coefficient is -0.2 , then the relationship between the variables is (A) low direct low inverse (C) high direct (D) high inverse
45. The sample size required to estimate the mean if the error is to be less than 1000 with 95% level confidence and standard deviation 10000 is (A) 384 (B) 19 (C) 20 (D) 385
46. A sample of 75 renters in a city showed that 40 were under the age of 35. Construct a 99% confidence interval for the percentage of all renters that are under the age of 35. (A) (0.38, 0.68) (C) (0.36, 0.97) (D) (0.42, 0.92)

47. Given that the calculated value for large sample one-tailed test at $\alpha = 1\%$ is $Z = -2.21$. The P-value for the test is -- (A) 0.03 (B) 0.01 (C) 0.97 (D) 0.99
48. The variance of a set of data is 1.2. Find the new standard deviation if each of the observation is multiplied by 2. (A) 4.80 (B) 2.40 (C) 2.19 (D) 1.55
49. The standard deviation of a set of data is 2.5. Find the new standard deviation if each of the observation is divided by 4. (A) 1.25 (B) 1.56 (C) 0.39 (D) 0.63
50. An estimate of population parameter that is based on sample is derived using sample -- (A) parameter (B) mean (C) statistic (D) variance
51. The sample size required for large sample inference is (A) $n \geq 30$ (B) $n \leq 30$ (C) $n \geq 100$ (D) $n \geq 30$
52. An estimator whose expected value equals the estimated value is said to be (A) unbiased (B) biased (C) correct (D) appropriate
53. The two conditions that lead to the use of small sample test statistic are (A) $n < 30$ and σ known (B) $n < 30$ and σ unknown (C) $n \geq 30$ and σ unknown (D) None of the above
54. Using the p-value, the null hypothesis is rejected if (A) $p < \alpha$ (B) $p > \alpha$ (C) $p = \alpha$ (D) $p \geq \alpha$
55. The statistical distribution appropriate for small sample test of significant difference in means is (A) Normal (B) Chi-Square (C) Student's t (D) Fisher F
56. Rejection of the null hypothesis when it is true leads to ----- error. (A) type II (B) one-tailed (C) type I (D) two-tailed
57. In test of hypothesis, the second step is (A) calculated value of the test statistic (B) H_0 and H_1 (C) critical region (D) decision
58. If $n = 144$, $\bar{x} = -2.1$, $\sigma = 1.2$. A 95% confidence interval for μ is ----- (A) (1.91, 2.30) (B) (-2.30, -1.91) (C) (-2.26, -1.94) (D) (1.94, 2.26)
59. If $n = 25$, $\bar{x} = -2.1$, $S = 1.1$. A 95% confidence interval for μ is ----- (A) (1.86, 2.34) (B) (-2.34, -1.86) (C) (1.60, 2.60) (D) (-2.60, -1.60)
- Use the following information to answer questions 61 to 62.
If $\bar{x}_1 = 6.5$, $S_1 = 2.5$, $n_1 = 100$, $\bar{x}_2 = 8.5$, $S_2 = 2.6$, $n_2 = 200$.
60. The value of the test statistic for testing the significant difference between the two sample means is ---- (A) -6.45 (B) 6.45 (C) -10.26 (D) 10.26
61. What is the decision for the hypothesis using $\alpha = 5\%$ (A) Accept H_0 (B) Cannot reject H_0 (C) Reject H_0 (D) Reserve judgment
- Use the following information to answer questions 63 to 66.
If $\bar{x}_1 = 5$, $S_1 = 1.2$, $n_1 = 25$, $\bar{x}_2 = 3$, $S_2 = 1.5$, $n_2 = 16$.
62. The pooled standard deviation for the two samples is (A) 1.32 (B) 1.75 (C) 4.73 (D) 3.57
63. The value of the test statistic for testing the significant difference between the two sample means is -- (A) 3.57 (B) 1.75 (C) 4.73 (D) 1.32
64. What is the table value for testing the hypothesis using $\alpha = 1\%$. (A) 1.685 (B) 2.023 (C) 2.426 (D) 2.708
65. What is the decision for the hypothesis using $\alpha = 1\%$ (A) Accept H_0 (B) Cannot reject H_0 (C) Reject H_0 (D) Reserve judgment
66. A bakery claims that the weight of its bread is more than 500g. To test if the actual weight is less than the company claim, samples were selected and hypothesis testing carried out. What type of test is this (A) type I (B) type II (C) two-tailed (D) one-tailed
- Use the following information to answer questions 68 to 70.
A 95% confidence interval reported for the parameter μ is 1.5 to 6.5. The interval is based on 15 observations.
67. What is the table value of the t-distribution? (A) 2.145 (B) 2.624 (C) 1.761 (D) 2.977
68. What is the value of the standard error of mean ($s.e(\bar{x})$) (A) 1.19 (B) 1.17 (C) 1.96 (D) 1.86
69. What is the mean of the unknown data set (A) 8.0 (B) 12.0 (C) 2.5 (D) 4.0
70. What is the mean of the unknown data set

No of hr	0-9	10-19	20-29	30-39	40-49	50-59
frequency	40	50	70	40	30	20

1) Mode is given by $L_1 + \left[\frac{D_1}{D_1 + D_2} \right] C$ where $D_1 = f_1 - f_0$
 $D_2 = f_1 - f_2$
 $C = \text{class size}$
 $L_1 = \text{Lower class limit of the modal class}$

$$\therefore \text{Mode} = 19.5 + \left[\frac{70-50}{(70-50)+(70-40)} \right] \times 10$$

$$= 19.5 + \left[\frac{20}{20+30} \right] \times 10 \Rightarrow 19.5 + 4 = \underline{\underline{23.5}} (D)$$

2) $Q_1 = L_1 + \left[\frac{\frac{N}{4} - Cfb}{f_{Q_1}} \right] \times C$ where $N = \sum f$
 $\frac{N}{4} = 1^{\text{st}} \text{ quartile position}$

$$\Rightarrow Q_1 = 9.5 + \left[\frac{\frac{250}{4} - 40}{50} \right] \times 10 = 9.5 + 4.5 = \underline{\underline{14}} (B)$$

3) Median = $L_1 + \left[\frac{\frac{N}{2} - Cfb}{f_m} \right] \times C$ $\frac{N}{2} \rightarrow \text{median position}$

$$\Rightarrow 19.5 + \left[\frac{\frac{250}{2} - 90}{70} \right] \times 10 \Rightarrow 19.5 + 5 = \underline{\underline{24.5}} (C)$$

4) $\bar{x} = \frac{\sum x}{N}$, $\frac{\sum x}{5} = 20 \Rightarrow \sum x = 100$

if all the numbers i.e. $x_1, \dots, x_n$ is multiplied by 2 and added by 5;
 Then $\sum x$ is multiplied by 2 and added by 25 since $n=5$.

$$\Rightarrow (100 \times 2) + 25 = 225$$

$$\bar{x} = \frac{225}{5} = \underline{\underline{45}} (D)$$

Renanced

$$5) \frac{1+2+3+4+5+6+7+8+9+10}{10} = \frac{55}{10} = \underline{\underline{5.5}} (B)$$

$$6) 11, 11, 12, 12, 13, 13, 13, 15, 18$$

$$\text{mode} = 13, \text{median} = 13, \text{mean} = \frac{118}{9} = 13.11$$

$$\therefore \text{mode} + \text{median} + \text{mean} = 13 + 13 + 13.11 = 39.11 \approx \underline{\underline{39}} (A)$$

7) (A) check number (5) above

$$8) \frac{(6-x) + (7+2x) + (3-3x) + (5+4x) + (4-5x) + (11+6x) + (2-7x) + (18+8x)}{8} = 10$$

$$\Rightarrow 56 + 4x = 80 \Rightarrow 4x = 24 \Rightarrow x = \underline{\underline{6}} (B)$$

$$9) G = \sqrt[n]{x_1 x_2 \dots x_n}$$

$$\therefore G = \sqrt[13]{4 \times 7 \times 3 \times 5 \times 3 \times 6 \times 4 \times 4 \times 5 \times 4 \times 6 \times 6 \times 8} = \underline{\underline{4.79}} (B)$$

Class Interval	x	f	fx	fx^2	C.f
13-17	15	2	30	450	2
18-22	20	22	440	8800	24
23-27	25	10	250	6250	34
28-32	30	14	420	12600	48
33-37	35	3	105	3675	51
38-42	40	4	160	6400	55
43-47	45	6	270	12150	61
48-52	50	1	50	2500	62
53-57	55	1	55	3025	63
Total		63	1780	55850	

$$10) S.D = \sqrt{\frac{\sum f x^2}{N} - \left(\frac{\sum f x}{N}\right)^2} \Rightarrow \sqrt{\frac{55850}{63} - \left(\frac{1780}{63}\right)^2}$$

$$S.D = \sqrt{886.5079 - 798.2867} = \underline{9.3926} \text{ (C)}$$

$$11) \text{Var} = (S.D)^2 \Rightarrow (9.3926)^2 = \underline{88.22} \text{ (A)}$$

$$12) CV = \frac{S.D}{\bar{x}} \times 100, \bar{x} = \frac{\sum f x}{\sum f} = \frac{1780}{63} = 28.25$$

$$\therefore CV = \frac{9.3926}{28.25} \times 100 = \underline{33.25} \text{ (A)}$$

$$13) \text{Degree of asymmetry \{Skewness\}} = \frac{\bar{x} - \text{mode}}{S.D}$$

$$\text{mode} = L_1 + \left[\frac{D_1}{D_1 + D_2} \right] \times C = 17.5 + \left[\frac{22-2}{(22-2)+(22-10)} \right] \times 5$$

$$\Rightarrow 17.5 + 3.125 = 20.625$$

$$\therefore \text{Skewness} = \frac{28.25 - 20.625}{9.3926} = \underline{0.81} \text{ (No option)}$$

$$14) \underline{61} \text{ (C)}$$

$$15) Q_3 \text{ position} = \frac{3N}{4} = \frac{3 \times 63}{4} = 47.25$$

The c.f before 47.25 is 34 (A)

$$16) H = \frac{N}{\sum \frac{1}{x}} = \frac{13}{\frac{1}{4} + \frac{1}{7} + \frac{1}{3} + \frac{1}{5} + \frac{1}{3} + \frac{1}{6} + \frac{1}{4} + \frac{1}{4} + \frac{1}{5} + \frac{1}{4} + \frac{1}{6} + \frac{1}{6} + \frac{1}{8}}$$

$$\Rightarrow \frac{13}{2381} = \underline{4.59} \text{ (B)}$$

40) (C)

41) (B)

42) (B)

43) (C)

44) (D)

45) $z_{95\%} = 1.96$, $\sigma = 10000$, $e < 1000 \therefore e = 999$

$$n = \frac{z^2 \times \sigma(1-\sigma)}{e^2} \Rightarrow \frac{(1.96)^2 \times 10000(1-10000)}{999^2}$$

 $n = -384.89$ Since n cannot be $-ve$, $n = \underline{\underline{385}}$ (D)

46) incomplete parameters

47) (A) Just look up the table value corresponding to -0.6

48) (B) same as No 47.

49) $\text{Var}(ax) = a^2 \times \text{Var}(x)$ where $a = 2$, $\text{Var}(x) = 1.2$
 $\therefore \text{Var}(2x) = 2^2 \times 1.2 = \underline{\underline{4.8}}$ (A)

50) (C)

50) $\sigma = \sqrt{\text{Var}} \Rightarrow \text{Var} = \sigma^2$ where $\sigma = 2.5$, $\text{Var} = 6.25$

$$\text{Var}\left(\frac{x}{a}\right) = \left(\frac{1}{a}\right)^2 \times \text{Var}(x) \text{ where } a = 4, \text{Var}(x) = 6.25$$

$$\therefore \text{Var}\left(\frac{x}{4}\right) = \left(\frac{1}{4}\right)^2 \times 6.25 = \underline{\underline{0.625}}$$
 (D)

51) (C)

52) (D)



53) (A)

54) (B)

55) (A)

56) (C)

57) (C)

58) (A)

$$59) n = 144, \bar{x} = -2.1, \sigma = 1.2, \alpha = 95\% = 0.95$$

$$Z_{\left(\frac{1-\alpha}{2}\right)} = Z_{\left(\frac{1-0.95}{2}\right)} = Z_{0.025} = 1.96$$

$$\mu = \bar{x} \pm Z_{\frac{\alpha}{2}} \times S.E \Rightarrow \mu = -2.1 \pm 1.96 \times \frac{1.2}{\sqrt{144}}$$

$$\Rightarrow \mu = -2.1 \pm 0.196 \Rightarrow \mu = (-2.296, -1.904) \text{ (B)}$$

$$60) n = 25, \bar{x} = -2.1, s = 1.1, \alpha = 95\% = 0.95, 1 - \alpha = 0.05$$

$$t_{\left(1-\frac{\alpha}{2}\right), n-1} = t_{(0.975), 24} = 2.064$$

$$\mu = \bar{x} \pm t_{\left(1-\frac{\alpha}{2}\right), n-1} \times S.E \Rightarrow \mu = \bar{x} \pm 2.064 \times \frac{s}{\sqrt{n}}$$

$$\mu = -2.1 \pm 2.064 \times \frac{1.1}{\sqrt{25}} \Rightarrow -2.1 \pm 0.45408$$

$$\mu = (-2.55, -1.65) \text{ (D)}$$

$$61) \bar{x}_1 = 6.5, s_1 = 2.5, n_1 = 100, \bar{x}_2 = 8.5, s_2 = 2.6, n_2 = 200$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2 - c}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \Rightarrow \frac{6.5 - 8.5 - 0}{\sqrt{\frac{2.5^2}{100} + \frac{2.6^2}{200}}} = \frac{-2}{0.3103} = -6$$

$$Z = -6.44 \text{ (A)}$$

Note: $c = 0$ when not stated or given

$$62) \alpha = 5\% = 0.05, Z_{(1-\frac{\alpha}{2})} = Z_{(1-\frac{0.05}{2})}$$

$$\Rightarrow Z_{0.975} = 1.96$$

Since the calculated value (-6.44) is less than the tab. value (1.96) at $\alpha = 5\%$, we Accept H_0 (A)

$$63) \bar{x}_1 = 5, s_1 = 1.2, n_1 = 25, \bar{x}_2 = 3, s_2 = 1.5, n_2 = 16$$

$$S_{pooled} = \sqrt{\frac{s_1^2 + s_2^2}{2}} \quad \text{or} \quad \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

$$\therefore S_{pooled} = \sqrt{\frac{1.2^2 + 1.5^2}{2}} \quad \text{or} \quad \sqrt{\frac{(25-1) \times 1.2^2 + (16-1) \times 1.5^2}{25+16-2}}$$

$$= \underline{1.3583} \quad \text{or} \quad \underline{1.3235 (A)}$$

$$64) t = \frac{\bar{x}_1 - \bar{x}_2 - c}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \Rightarrow \frac{5-3-0}{1.3235 \sqrt{\frac{1}{25} + \frac{1}{16}}}$$

$$t = \underline{4.72 (C)}$$

$$65) \text{ at } \alpha = 1\% = 0.01, t_{(1-\frac{\alpha}{2}), n_1+n_2-2}$$

$$\Rightarrow t_{(1-\frac{0.01}{2}), 25+16-2} \Rightarrow t_{0.995, 39} = \underline{2.708 (D)}$$

66) since $t_{cal} > t_{tab}$, Reject H_0 (C)

67) (D)

$$68) \alpha = 95\% = 0.95, 1 - \alpha = 0.05, n = 15$$

$$t_{(1-\frac{\alpha}{2}), n-1} \Rightarrow t_{(1-\frac{0.05}{2}), 15-1} \Rightarrow t_{0.975, 14}$$

$$\Rightarrow t_{0.975, 14} \Rightarrow \underline{\underline{2.145}} \text{ A}$$

$$69) \mu = \bar{x} \pm t_{\frac{\alpha}{2}} \times \text{s.e.}, \mu = 1.5, \text{ to } 6.5$$

$$\therefore 1.5 = \bar{x} - 2.145 \text{ s.e.} \quad \text{--- (1)}$$

$$6.5 = \bar{x} + 2.145 \text{ s.e.} \quad \text{--- (2)}$$

Subtract eqn(1) from eqn(2)

$$5 = 4.29 \text{ s.e.} \Rightarrow \text{s.e.} = \frac{5}{4.29} = \underline{\underline{1.166}} \text{ (B)}$$

70) add eqn(1) to eqn(2)

$$8 = 2\bar{x} \Rightarrow \bar{x} = \frac{8}{2} = \underline{\underline{4}} \text{ (D)}$$



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2016/2017
Course Code: MTH 112
Level: 100

Name: Title: Data Analysis
Duration: 2½ hrs
Matric No: S/N on Attendance:
Instruction: Answer all questions by shading the correct option in the question paper and copy the same option on the first page of the booklet. Submit both the question paper and the booklet for marking.

TYPE A

- Rejection of the null hypothesis when it is true leads to error.
(a) type I (b) type II (c) one-tailed (d) two-tailed
- The critical region is the region set aside for
(a) accepting H_0 (b) rejecting or accepting H_0 (c) rejecting H_0 (d) reserving judgment
- The critical value for the rejection of the null hypothesis for large sample one-tailed test at $\alpha = 1\%$ is
(a) 2.56 (b) 1.64 (c) 1.64 (d) 2.33
- If $n = 225$, $\bar{x} = -5.2$, $\sigma = 2.5$. A 99% confidence interval for μ is
(a) (4.77, 5.63) (b) (-5.29, -4.91) (c) (-5.63, -4.77) (d) (-6.91, -7.29)
Use the following information to answer questions 5 to 7.
If $\bar{x}_1 = 10.5$, $S_1 = 12.5$, $n_1 = 144$, $\bar{x}_2 = 8.5$, $S_2 = 11.5$, $n_2 = 100$.
- The calculated value of the test statistic for testing the significant difference between the two sample means is
(a) 1.96 (b) 1.29 (c) 5.62 (d) 2.58
- What is the decision for the hypothesis using $\alpha = 5\%$
(a) Accept H_0 (b) Reject H_0 (c) Reserve judgment (d) None of the above
- The P-value for the hypothesis using $\alpha = 5\%$ is
(a) 1.96 (b) 0.582 (c) 0.197 (d) 1.64
Use the following information to answer questions 8 to 10.
If $\bar{x}_1 = 5$, $S_1 = 0.5$, $n_1 = 10$, $\bar{x}_2 = 5.5$, $S_2 = 0.25$, $n_2 = 15$.
- The calculated value of the test statistic for testing the significant difference between the two sample means is
(a) 1.96 (b) 2.36 (c) 5.36 (d) 3.32
- What is the table value for testing the hypothesis using $\alpha = 1\%$
(a) 2.807 (b) 2.797 (c) 2.500 (d) 2.492
- The hypothesis decision is (a) Reserve judgment (b) Accept H_0 (c) Reject H_0 (d) None of the above
- If $n = 25$, $\bar{x} = 15.15$, $S = 1.5$. A 95% confidence interval for μ is
(a) (14.56, 15.74) (b) (14.53, 15.77) (c) (6.51, 7.69) (d) (16.69, 17.51)
- A company claims that its bag of rice weight 50kg. To test if the actual weight is less than the company claim, samples were selected and hypothesis testing carried out. What type of test is this
(a) type I (b) type II (c) two-tailed (d) one-tailed
- H_0 for testing hypothesis using normal distribution is rejected if
(a) $Z_{\text{cal}} < Z_{\text{tab}}$ (b) $P\text{-value} < \alpha$ (c) $Z_{\text{cal}} \leq Z_{\text{tab}}$ (d) $P\text{-value} > \alpha$
- If $\alpha = 7.5\%$, what is the table value for a two-tailed test using normal distribution
(a) 1.98 (b) 1.64 (c) 1.78 (d) 1.76
- Determine the P-value for a one-tailed test if the calculated value of the normal distribution statistic is -1.72 (a) 0.0427 (b) 1.0962 (c) 0.9573 (d) 1.0952
- The time taken by a typist to complete a task on a machine is normally distributed with $\sigma = 2.5$ mins. In a sample of 25 tasks, $\bar{x} = 25.5$ mins was found. Construct a 95% confidence interval for μ
(a) (25.64, 26.96) (b) (24.52, 26.48) (c) (21.64, 26.96) (d) (24.52, 26.75)
- The 99% confidence interval for μ given a data set with $n = 36$, $\bar{x} = 15.5$, $S = 0.75$ is
(a) (13.50, 15.58) (b) (15.50, 15.58) (c) (15.18, 15.82) (d) (15.40, 15.50)
- Use the following information to answer questions 18 to 20. A 95% confidence interval reported for the parameter μ is 5.27 to 14.73. The interval is based on 5 observations constructed using t-distribution.
(a) 2.131 (b) 2.776 (c) 2.582 (d) 1.642
- What is the table value of the t-distribution
(a) 6.80 (b) 6.48 (c) 1.70 (d) 4.50
- What is the value of the standard error of mean (s.e($\bar{x}$))
(a) 10.00 (b) 11.50 (c) 12.50 (d) 20.50
- What is the mean of the unknown data set
(a) 10.00 (b) 11.50 (c) 12.50 (d) 20.50
- Use the following information to answer questions 21 to 23:
The mean score in an examination from a random sample of 25 students is 47.50 with $S = 2.50$.
- The calculated value of the statistic for testing the claim that the mean score is higher than 45 at $\alpha = 1\%$
(a) 2.81 (b) 2.58 (c) 5.64 (d) 5.00
- The table value for the test is
(a) 1.96 (b) 2.49 (c) 1.64 (d) 2.58

tof A

23. The hypothesis decision is (a) Reserve judgment (b) Accept H_0 (c) Reject H_0 (d) None of the above
24. The variance of a set of data represented by Y is 1.50 then the variance of $(Y - 2)$ is (a) 6.0 (b) 1.5 (c) 3.0 (d) 4.5
25. X represents a set of data with variance 5.0, the variance of $\left(\frac{X}{2}\right)$ is (a) 1.25 (b) 2.50 (c) 5.0 (d) 7.5
- Use the following information to answer questions 26 to 30. A random sample of 36 Shoprite customers spent mean amount of 450.50 naira and standard deviation of 100.50 naira.
26. The standard error of means for this sample is (a) 6.75 (b) 1.96 (c) 1.65 (d) 16.75
27. The calculated value of the statistic for testing the claim that the mean amount spent is less than 400 at 0.05 level of significance is (a) 1.03 (b) 3.01 (c) 3.50 (d) 1.11
28. The table value for this test is (a) 1.96 (b) 2.33 (c) 1.65 (d) 2.58
29. The P-value for this test is (a) 0.0013 (b) 2.0313 (c) 0.6523 (d) 2.5823
30. The hypothesis decision is (a) Reserve judgment (b) Reject H_0 (c) Accept H_0 (d) None of the above
- Use the following information to answer questions 31 to 35. The following are the test scores of students 30, 16, 10, 24.
31. The unbiased estimate of the population standard deviation is (a) 6.81 (b) 11.60 (c) 3.81 (d) 7.62
32. The standard error of mean for the sample is (a) 6.80 (b) 3.05 (c) 3.41 (d) 4.50
33. The calculated value of the statistic for testing the claim that the sample mean is not significantly different from at $\alpha = 5\%$ is (a) -0.59 (b) 3.01 (c) -3.50 (d) 0.59
34. The critical value used for the test in question 33 is (a) 1.96 (b) 2.58 (c) 2.78 (d) 1.64
35. The hypothesis decision is (a) Reject H_0 (b) Accept H_0 (c) Reserve judgement (d) None of the above
36. Let $\{(1,5), (3,4), (m,2), (4,5), (5,1), (1,3), (1,4), (2,5), (3,5), (0,6)\}$ be points on the Euclidean plane with equal means. Find the sample variance of x. (a) 20 (b) 19 (c) 34 (d) 6
37. Find the value of $\text{var}(X - Y)$ if the $\text{var}(X)$, $\text{var}(Y)$ and $\text{cov}(X, Y)$ are $2\frac{1}{2}$, 4, and $\frac{1}{2}$ respectively. (a) 7 (b) $6\frac{1}{2}$ (c) 3 (d) 6
38. The test records for mathematics and computer students are presented in pair respectively (1,2), (3,3), (3,5), (5,6), (6,5), (8,7), (9,8). Find the $\text{cov}(X, Y)$. (a) 3.50 (b) 6.25 (c) 4.24 (d) 5.00
39. Find the coefficient of variation for computer students of test records for Mathematics and Computer students (3,3), (3,5), (5,4), (5,6), (6,5), (8,7), (9,8). (a) 50 (b) 70 (c) 37 (d) 125
40. Find the $\text{cov}(20, a)$ if a is a constant of variation whose probability of occurrence is one. (a) 20 (b) 400 (c) 0 (d) a
41. If the $\text{var}(X) = 15$. Find the $\text{cov}(aX, X)$ if X is a random variable of a population and $a = 2$ (a) 15 (b) 225 (c) 15a (d) 15a
42. What is the value of x if the $2x - 1$ is the sectorial angle of $1/12$ with a pie-chart whole cumulative is 12 (a) 21 (b) 14 (c) $1\frac{1}{2}$ (d) 12
43. Find the $\text{var}(X - Y)$ if X and Y are independently and identically distributed if the $\text{var}(X)$, $\text{var}(Y)$ and $\text{cov}(X, Y)$ are $2\frac{1}{2}$, 4, and $\frac{1}{2}$ respectively. (a) 6 (b) 7 (c) $6\frac{1}{2}$ (d) 3
44. When frequency data and count data are represented by counts or number of observations in each group observation in a population, they are referred to as --- data (a) categorical (b) measurement (c) quantitative (d) parameter
45. --- assigns numbers to object of a sample in an experiment. (a) measurement (b) parameter (c) statistical (d) nominal scale
46. Find the mean of the distribution 66, 75, 72, 71, 55, 56, 72, 93, 72, 72, 73, 73, 91, 66, 71, 56, 59. (a) 66 (b) 72.00 (c) 66.3 (d) 70
47. Which of the following represent the characteristics of a good questionnaire? (a) Tele-guiding questions (b) Long and explanatory questions (c) Honest and yogue questions (d) Unambiguous

The table below represents the examination scores of 100 levels students in MTH 112 in 2013/2014 session mean 18.38. Use the table to answer questions 48 to 50

x	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30
f	y	7	11	10	6

48. Find the variance of the distribution (a) 6.00 (b) 18.38 (c) 41.52 (d) 6.40
49. Find the value of y in the distribution (a) 6.0 (b) 18.00 (c) 41.00 (d) 8.00
50. What is the median of the distribution (a) 6.0 (b) 18.3 (c) 41.00 (d) 21.00
51. Find the mode of 66, 75, 72, 71, 55, 91, 66, 71, 56, 59, 56, 72, 93, 72, 72, 73, 73 (a) 66 (b) 72 (c) 56 (d) 73

Use the information to answer questions 52 - 55

x	1.2	1.3	1.4	2.4	0.8	3.4	0.2	1
y	5	6	1	2.9	4.2	k	5	3

52. What is the value of k if the slope is 1.327? (a) 7 (b) 3 (c) 10 (d) 8
53. Find the intercept if $y = a + bx$ is the regression equation (a) 3 (b) 7 (c) 8 (d) 10

2 of A

54. Which of the equation represents the regression equation? (a) $y = 3 + 1.3x$ (b) $y = 7 + 1.8x$ (c) $y = 3 - 1.8x$ (d) $y = 4.2 + 8x$
55. What is the mean of the variable y ? (a) 4.64 (b) 1.46 (c) 10.00 (d) 1.00
56. What is the value of y if the mean of the number $13, 2y + 1, 0, 5y - 1$, and 11 is 9 ? (a) 1.7 (b) 3.0 (c) 4.0 (d) 7.0
57. The mean and the median of a symmetric distribution is said to be equal when they have ----- distribution? (a) unimodal (b) bimodal (c) multimodal (d) mode
58. The staff strength of Jom-oil Company is shown below. Find the median.
- | | | | | | | | | | | |
|----------------|------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| Staff strength | 1-10 | 11-20 | 21-30 | 31-40 | 41-50 | 51-60 | 61-70 | 71-80 | 81-90 | 91-100 |
| No. of company | 1 | 5 | 10 | 19 | 42 | 10 | 6 | 4 | 2 | 1 |
- (a) 44.07 (b) 42.00 (c) 77.00 (d) 45.00
59. The time taken in seconds by 100 different chemical substances to melt when subjected to a particular temporary condition are given below. Find the mode
- | | | | | | | |
|------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Time (sec) | 4.51 - 5.32 | 5.33 - 6.14 | 6.15 - 6.96 | 6.97 - 7.78 | 7.79 - 8.60 | 8.61 - 9.42 |
| frequency | 15 | 7 | 35 | 28 | 10 | 5 |
- (a) 6.145 (b) 6.801 (c) 6.965 (d) 35
60. Find the probability of the class interval 30 - 39 in the table below
- | | | | | | | |
|---|-----|-------|-------|-------|-------|-------|
| x | 0-9 | 10-19 | 20-29 | 30-39 | 40-49 | 50-59 |
| y | 10 | 2 | 4 | 5 | 25 | 14 |
- (a) 1 (b) $3/10$ (c) $1/5$ (d) $1/12$
61. What is the standard deviation of the distribution for an average of 48.28?
- | | | | | | |
|---|----|----|---------|----|----|
| X | 25 | 29 | $y + 3$ | 59 | 70 |
| f | 7 | 9 | 8 | 6 | 20 |
- (a) 366.96 (b) 18.96 (c) 359.63 (d) 19.16
62. Find the value of y for an average of 48.28 in the table below
- | | | | | | |
|---|---------|----|----|----|----|
| X | $y + 3$ | 70 | 25 | 59 | 29 |
| f | 8 | 20 | 7 | 6 | 9 |
- (a) 38 (b) 50 (c) 40 (d) 25
63. Given the table below for 100 level 50 test scripts
- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 11-15 | 16-20 | 21-25 | 26-30 | 31-35 | 36-40 |
| 6 | 5 | x | 12 | 9 | 5 |
- If the class intervals are represented in a pie chart, what is the sectorial angle of the class interval 26-30? (a) 43.2 (b) 86.4 (c) 93.6 (d) 64.8
64. Find the class size of this distribution
- | | | | | |
|-----------|-------|-------|-------|-------|
| x | 26-30 | 31-35 | 36-40 | 41-45 |
| frequency | p | y | 10 | 21 |
- (a) 15.5 (b) 5.0 (c) 25.6 (d) 30.5
65. The normal distribution is represented by
- (a) $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ (b) $\frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2}}$ (c) $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ (d) $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$
- Use the information below to answer question 66 and 67
- | | | | | | |
|---|-------|-------|-------|-------|-------|
| x | 15-19 | 20-24 | 25-29 | 30-34 | 35-40 |
| f | 3 | 4 | 7 | 5 | 1 |
66. Find the mean of the distribution (a) 27.50 (b) 26.25 (c) 7.00 (d) 30.00
67. Find the skewness (a) 0.40 (b) 2.25 (c) 0.22 (d) -0.40
68. The table below shows a frequency distribution for grouped data. Calculate the mode
- | | | | | | | |
|-----------|-----|------|-------|-------|-------|--------|
| x | 0-5 | 5-15 | 15-25 | 25-35 | 35-45 | 45-100 |
| frequency | 2 | 6 | 20 | 15 | 8 | 4 |
- (a) 24.75 (b) 22.37 (c) 27.73 (d) 27.50
69. Find the sum of the deviation from the mean of the distribution 4, 6, 2, 8, 10, 3, 6, 4, 7, 10. (a) 22 (b) 6 (c) 0 (d) 7.8
70. ----- is a numerical value that summarizes population
- (a) Parameter (b) Statistic (c) Inferential statistic (d) Descriptive statistic

Best of Luck.

Examiners: Dr. A. M. Okedoye, S. O. Ogundele,

J. T. Eghwerido.

MTH112 (2016/2017) EXAM TYPE A

1) A

2) C

$$3) Z_{Tab} = Z_{(1-\frac{\alpha}{2})} \Rightarrow Z_{(1-\frac{1}{2 \times 100})} = Z_{0.995} = \underline{2.58} \quad (B)$$

$$4) n = 225, \bar{x} = -5.2, \sigma = 2.5, \alpha = 99\% = 0.99 \{1 - 0.99 = 0.01\}$$

$$\mu = \bar{x} \pm \left\{ Z_{(1-\frac{\alpha}{2})} \times S.E. \right\} \text{ or } \mu = \bar{x} \pm \left[Z_{(1-\frac{\alpha}{2})} \times \frac{\sigma}{\sqrt{n}} \right]$$

$$\therefore \mu = -5.2 \pm \left[Z_{(1-\frac{0.01}{2})} \times \frac{2.5}{\sqrt{225}} \right] = -5.2 \pm [2.58 \times 0.167]$$

$$\therefore \mu = -5.2 \pm 0.43086 = \underline{(-5.63, -4.77)} \quad (C)$$

$$5) \bar{x}_1 = 10.5, s_1 = 12.5, n_1 = 144, \bar{x}_2 = 8.5, s_2 = 11.5, n_2 = 100$$

$$Z_{cal} = \frac{\bar{x}_1 - \bar{x}_2 - C}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{10.5 - 8.5 - 0}{\sqrt{\frac{12.5^2}{144} + \frac{11.5^2}{100}}} \quad \text{Note: } C = 0 \text{ if not given!}$$

$$Z = \underline{1.29} \quad (B)$$

$$6) Z_{Tab} = Z_{(1-\frac{\alpha}{2})} = Z_{(1-\frac{5}{100} \times \frac{1}{2})} = Z_{(0.975)} = 1.96$$

Since Z_{cal} is less than Z_{Tab} we Accept H_0 (A)

7) (B)

8) $\bar{x}_1 = 5, S_1 = 0.5, n_1 = 10, \bar{x}_2 = 5.5, S_2 = 0.25, n_2 = 15$
 note $C = 0$ {not given}

$$t_{cal} = \frac{\bar{x}_1 - \bar{x}_2 - C}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

but $S_p = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$

$$\therefore S_p^2 = \frac{(10 - 1) \times 0.5^2 + (15 - 1) \times 0.25^2}{10 + 15 - 2} = 0.1359$$

$$\therefore t_{cal} = \frac{5 - 5.5 - 0}{\sqrt{0.1359} \left(\sqrt{\frac{1}{10} + \frac{1}{15}} \right)}$$

note $S_p^2 = 0.1359$
 $\therefore S_p = \sqrt{0.1359}$

$$t_{cal} = 5.3227 \quad (D)$$

9) $t_{Tab} = t_{(1-\frac{\alpha}{2}), (n_1 + n_2 - 2)}$ but $\alpha = 1\% = 0.01$

$$t_{Tab} = t_{(1-\frac{0.01}{2}), (10+15-2)} = t_{(0.995), (23)} = 2.807 \quad (A)$$

10) Since $t_{cal} > t_{Tab}$ Reject H_0 (C)

11) $n = 25, \bar{x} = 15.15, S = 1.5, \alpha = 95\% = 0.95, \{1 - 0.95 = 0.05\}$

$$\mu = \bar{x} \pm \left[Z_{(1-\frac{\alpha}{2})} \times S.E \right]$$

$$\mu = 15.15 \pm \left[1.96 \times \frac{1.5}{\sqrt{25}} \right] \quad \text{please see question 4 for more details}$$

$$\mu = 15.15 \pm 0.588$$

$$\mu = 14.562, 15.738 \quad (A)$$

12) (D) note: we are only concerned with less than (i.e. left sided)

13) H_0 is rejected if P-Value $< \alpha$ (B)

$$14) Z_{\text{tab}} = Z_{(1-\frac{\alpha}{2})} \Rightarrow Z_{(1-\frac{0.05}{2})} = Z_{(0.975)} = 1.96 \text{ (C)}$$

15) (A)

16) $n=25, \bar{x}=25.5, \sigma=2.5, \alpha=95\%=0.05$ (see Q11 and 4)
 $\mu = (24.52, 26.48)$ (B)

17) $n=25, \bar{x}=15.5, s=0.75, \alpha=99\%=0.01$ (see Q4)
 $\mu = (15.18, 15.82)$ (C)

18) Given: $\alpha=95\%=0.05, \mu_u=14.73, \mu_l=5.27$
 $n=5$

19) $t_{\text{tab}} = t_{(1-\frac{\alpha}{2}), (n-1)} \Rightarrow t_{(1-\frac{0.05}{2}), (5-1)} = t_{(0.975), (4)} = 2.776$

$t_{\text{tab}} = 2.776$ (B)

19) Recall $\mu = \bar{x} \pm t_{(1-\frac{\alpha}{2}), (n-1)} \times S.E(\bar{x})$, but $t_{(1-\frac{\alpha}{2}), (n-1)} = t_{\text{tab}}$

$\mu_l = 5.27 = \bar{x} - 2.776 S.E(\bar{x})$ — (1)

$\mu_u = 14.73 = \bar{x} + 2.776 S.E(\bar{x})$ — (2) add eq (1) and (2)

$20 = 2\bar{x}$

$\bar{x} = 10$ — (*)

(Q19 Cont.)

Sub eqn (*) into eqn (1)

$$5.27 = \bar{x} - 2.776 S.E(\bar{x}) \quad (1), \bar{x} = 10 \quad (*)$$

$$\Rightarrow 5.27 = 10 - 2.776 S.E(\bar{x})$$

$$\therefore 2.776 S.E(\bar{x}) = 10 - 5$$

$$S.E(\bar{x}) = \frac{5}{2.776} = \underline{\underline{1.80}} \quad (C)$$

20) From eqn (*) $\bar{x} = \underline{\underline{10}} \quad (A)$ Given: $n = 25$, $\bar{x} = 47.5$, $S = 2.5$, $\mu_0 = 45$, $\alpha = 1\% = 0.01$

$$21) t_{cal} = \frac{\bar{x} - \mu_0}{S.E(\bar{x})} = \frac{\bar{x} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{47.5 - 45}{\frac{2.5}{\sqrt{25}}}$$

$$\Rightarrow t_{cal} = \underline{\underline{5}} \quad (D)$$

$$22) t_{tab} = t_{(1-\frac{\alpha}{2}), (n-1)} \Rightarrow t_{(1-\frac{0.01}{2}), (25-1)} \Rightarrow t_{(0.995), (24)}$$

$$\Rightarrow t_{tab} = \underline{\underline{2.492}} \quad (\text{No answer}) \quad (B) \quad \text{note: } (1-\frac{\alpha}{2}) \text{ For one tailed test.}$$

23) Reject $H_0 \quad (C)$ 24) (B) note $\text{Var}(x+a) = \text{Var}(x)$ where "a" is a constant.

$$25) V[nx] = n^2 \cdot V[x] \quad \text{where } V[x] = 5, n = \frac{1}{2}$$

$$\therefore V\left[\frac{x}{2}\right] = \left(\frac{1}{2}\right)^2 \times 5 = \frac{5}{4} = \underline{\underline{1.25}} \quad (A)$$

Please note that Variance $[V]$ is mostly written as (Var)

Given: $n=36$, $\bar{x}=450.5$, $\sigma=100.5$

$$26) S.e = \frac{\sigma}{\sqrt{n}} = \frac{100.5}{\sqrt{36}} = \underline{16.75} \text{ (D)}$$

$$27) Z_{cal} = \frac{\bar{x} - \mu_0}{S.e} \quad \text{where } \mu_0 = 400$$

$$Z_{cal} = \frac{450.5 - 400}{16.75} = \underline{3.015} \text{ (B)}$$

$$28) Z_{tab} = Z_{(1-\frac{\alpha}{2})} \quad \text{but } \alpha = 0.05$$

$$\therefore Z_{tab} = Z_{(1-\frac{0.05}{2})} = Z_{0.975} = \underline{1.96} \text{ (A)}$$

29) (A)

30) (b)

$$31) \text{Standard deviation } (\sigma)^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$\text{but } \bar{x} = \frac{\sum x}{n} = \frac{30+20+16+10+24}{5} = 20$$

$$\therefore \sigma^2 = \frac{(30-20)^2 + (20-20)^2 + (16-20)^2 + (10-20)^2 + (24-20)^2}{5-1}$$

$$\sigma^2 = 58 \quad \therefore \sigma = \sqrt{58} = \underline{7.6158} \text{ (D)}$$

$$32) S.e = \frac{\sigma}{\sqrt{n}} = \frac{7.6158}{\sqrt{5}} = \underline{3.4059} \text{ (C)}$$

$$33) t_{cal} = \frac{\bar{x} - \mu_0}{S.e} = \frac{20-22}{3.4059} \quad \text{where } \mu_0 = 22$$

$$t_{cal} = \underline{-0.5872} \text{ (A)}$$

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(5)

$$34) \text{ Critical value} = t_{\text{tab}} = t_{(1-\frac{\alpha}{2}), (n-1)}$$

$$t_{\text{tab}} = t_{(0.975), (4)} = \underline{2.776} \text{ (C)}$$

$$35) \text{ Since } t_{\text{cal}} < t_{\text{tab}} \text{ Accept } H_0 \text{ (B)}$$

$$36) \bar{x} = \bar{y}$$

$$\bar{y} = \frac{5+4+2+5+1+3+4+5+3+6}{10} = \frac{38}{10} = 3.8$$

$$\therefore \bar{x} = 3.8 = \frac{1+3+m+4+5+1+1+2+3+0}{10}$$

$$\Rightarrow 3.8 = \frac{20+m}{10} \Rightarrow m = 18$$

$$\text{Variance of } x = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$\therefore \text{Var}(x) = \frac{(1-3.8)^2 + (3-3.8)^2 + (18-3.8)^2 + \dots + (0-3.8)^2}{10}$$

$$\text{Var}(x) = \frac{192.8}{10} = \underline{19.28} \text{ (D)}$$

$$37) \text{Var}(x-y) = \text{Var}(x) + \text{Var}(y) - 2\text{Cov}(x, y) \quad \{\text{for independent variables}\}$$

$$\text{Var}(x-y) = \text{Var}(x) + \text{Var}(y) \quad \{\text{for independent variables}\}$$

$$\text{where } \text{Var}(x) = 2\frac{1}{2}, \text{Var}(y) = 4, \text{Cov}(x, y) = \frac{1}{2}$$

$$\text{Var}(x-y) = 2\frac{1}{2} + 4 = \underline{6\frac{1}{2}} \text{ (B)}$$

$$38) \text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\text{but } \bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{1+3+3+5+5+6+8+9}{8} = 5$$

and

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{2+3+5+4+6+5+7+8}{8} = 5$$

(Q38 cont.)

$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1-5 = -4	2-5 = -3	12
3-5 = -2	3-5 = -2	4
3-5 = -2	5-5 = 0	0
5-5 = 0	4-5 = -1	0
5-5 = 0	6-5 = 1	0
6-5 = 1	5-5 = 0	0
8-5 = 3	7-5 = 2	6
9-5 = 4	8-5 = 3	12
		<u>34</u>

$$\therefore \text{Cov}(x, y) = \frac{34}{8-1} = \underline{4.86} \text{ (D)}$$

39) Coefficient of Variation (CV) = $\frac{s}{\bar{x}} \times 100$

$$\therefore \text{C.V of Comp. students} = \frac{\sigma_y}{\bar{y}}$$

$$\text{but } \sigma_y = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{-3^2 + (-2)^2 + \dots + 3^2}{8-1}}$$

pls see the Table above

$$\sigma_y = \sqrt{\frac{28}{7}} = 2$$

$$\therefore \text{C.V of Comp. students} = \frac{2}{5} \times 100 = \underline{40} \text{ (No answer)}$$

40) (D)

$$41) \text{Cov}(ax, by) = ab \text{Cov}(x, y) \therefore \text{Cov}(ax, x) = \boxed{a \text{Cov}(x, x)}$$

$$\text{but } \text{Cov}(x, x) = \text{Var}(\bar{x}), \text{ where } \text{Var}(\bar{x}) = \frac{\text{Var}(x)}{n}$$

{ Since x is "a" random variable, $n=1$ }

$$\therefore a \text{Cov}(x, x) = a \times \text{Var}(x) = a \times 15 \Rightarrow 2 \times 15 = \underline{30} \text{ (C)}$$

Signature

$$42) \frac{2x-1}{12} = \frac{1}{12} \Rightarrow 2x-1=1 \Rightarrow 2x=2 \\ \Rightarrow x = \underline{\underline{1}} \text{ (B)}$$

$$43) \text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y) \quad \{\text{for independent variables}\} \\ \therefore \text{Var}(X-Y) = 2\frac{1}{2} + 4 = \underline{\underline{6\frac{1}{2}}} \text{ (C)}$$

$$44) \text{ (C)}$$

$$45) \text{ (A)}$$

$$46) \bar{x} = \frac{\sum_i^n x_i}{n} = \frac{66+75+72+71+55+56+72+93+72+72+73+91+66+71+56+59}{17}$$

$$\bar{x} = \frac{1193}{17} = \underline{\underline{70.18}} \text{ (D)}$$

$$47) \text{ (D)}$$

X	x	f	fx	(x-x) ²	C.f	Class boundary
6-10	8	4=6	84	107.74	6	5.5 - 10.5
11-15	13	7	91	28.94	13	10.5 - 15.5
16-20	18	11	198	14	24	15.5 - 20.5
21-25	23	10	230	34	34	20.5 - 25.5
26-30	28	6	168	54	40	25.5 - 30.5
		40	250.72			

$$\bar{x} = \frac{\sum fx}{\sum f} = 18.38 \therefore \frac{687+8y}{34+y} = 18.38$$

$$\Rightarrow 687+8y = 624.92 + 18.38y$$

$$\Rightarrow 62.08 = 10.38y \Rightarrow y = 6$$

$$48) \text{ Variance} = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{250.72}{40 - 1} = \underline{\underline{6.43}} \text{ (D)}$$

$$49) y = \underline{\underline{6}} \text{ (A)}$$

$$50) \text{ median position} = \frac{n}{2} = \frac{40}{2} = 20 \quad \left\{ \begin{array}{l} \text{check the cumulative} \\ \text{frequency column to} \\ \text{see when 20 falls} \end{array} \right\}$$

$$\text{Median} = L + \left[\frac{\frac{n}{2} - C_{fb}}{f_m} \right] \times C$$

$$\therefore \text{median} = 15.5 + \left[\frac{\frac{40}{2} - 13}{11} \right] \times 5 = \underline{\underline{18.68}} \text{ (B)}$$

$$51) \text{ mode is the most occurring element} = \underline{\underline{72}} \text{ (B)}$$

$$52) \text{ From } y = a + bx \text{ where } b = \text{slope} = 1.327$$

$$\text{where } y = 5, x = 1.2$$

$$\therefore 5 = a + 1.327(1.2) \Rightarrow a = 3.4 \approx 3$$

$$\therefore \text{when } y = 14, x = 3.4$$

$$14 = 3.4 + (1.327 \times 3.4) = \underline{\underline{7.9}} \text{ (D)}$$

$$53) a = 3 \text{ (A)}$$

$$54) y = 3 + 1.3x \text{ (A)}$$

$$55) \bar{y} = \frac{\sum y_i}{n} = \frac{5+6+1+2.9+4.2+7.9+5+3}{8} = \underline{\underline{4.39}} \text{ (A)}$$

$$56) \frac{13 + (2y+1) + 0 + (5y-1) + 11}{5} = 9$$

$$\Rightarrow 7y + 24 = 45$$

$$y = \underline{\underline{3}} \text{ (B)}$$

57)

$$58) \text{ median position} = \frac{n}{2} = \frac{100}{2} = 50$$

$$\text{median} = L + \left[\frac{\frac{n}{2} - Cfb}{f_m} \right] \times C$$

where L is lower class boundary of the median class

Cfb is cumulative frequency before the median class

f_m is frequency of the class

C is class size

$$\Rightarrow 40.5 + \left[\frac{50 - 35}{42} \right] \times 10$$

$$\Rightarrow \underline{44.07} \text{ (A)} \quad \left\{ \text{Pls see Q 48 for example on } \right. \\ \left. \text{C.f and class Boundary Table} \right\}$$

$$59) \text{ Mode} = L + \left[\frac{D_1}{D_1 + D_2} \right] \times C \quad \text{where "L" is the lower class boundary of the modal class}$$

$$= 5.65 + \left[\frac{35 - 7}{(35 - 28) + (35 - 7)} \right] \times 0.82$$

$$D_1 = f_m - f_B$$

$$D_2 = f_m - f_A$$

$$\Rightarrow \underline{6.801} \text{ (B)}$$

$$60) \text{ probability} = \frac{\text{Required outcome}}{\text{Total possible outcome}}$$

$$\therefore P_{(30-39)} = \frac{5}{10+2+4+5+25+14} = \frac{5}{60}$$

$$= \underline{\underline{\frac{1}{12}}} \text{ (D)}$$

61)

x	25	29	y+3=28	59	70	Total(Σ)
f	7	9	8	6	20	50
fx	175	261	8y+24	354	1400	2214+8y
(x- $\bar{x}$) ²	541.96	371.72	411.28	114.92	471.76	1911.64

$$\bar{x} = 48.28 = \frac{\Sigma fx}{\Sigma f} = \frac{2214+8y}{50} \Rightarrow \frac{2214+8y}{50} = 48.28$$

$$8y = (50 \times 48.28) - 2214 \Rightarrow 8y = 200, \underline{y = 25}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma (x-\bar{x})^2}{\Sigma f}} = \sqrt{\frac{1911.64}{50}}$$

$$= \underline{6.18} \text{ (No answer)}$$

62) $y = \underline{25}$ (D) { see the solution above }

63) $\frac{12}{50} \times 360 = \underline{86.4^\circ}$ (B) {

64) class size = upper class boundary - lower class Boundary

$$\therefore C = 30.5 - 25.5 \text{ or } 45.5 - 40.5$$

$$= \underline{5} \text{ (B)}$$

65) Normal distribution $\Rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ (C)

x	15-19	20-24	25-29	30-34	35-40	Total
x	17	22	27	32	37	
f	3	4	7	5	1	20
fx	51	88	189	160	37	525
$(x - \bar{x})^2$	85.56	18.06	0.56	33.06	115.56	252.80

$$66) \bar{x} = \frac{\sum fx}{\sum f} = \frac{525}{20} = \underline{\underline{26.25}} (B)$$

$$67) \text{Skewness} = \frac{\text{mean} - \text{mode}}{\text{Standard deviation}}$$

$$\text{but mode} = L + \left[\frac{D_1}{D_1 + D_2} \right] \times C$$

$$\Rightarrow 24.5 + \left[\frac{7-4}{(7-4) + (7-5)} \right] \times 5 = 27.5$$

$$\text{and S.d} = \sqrt{\frac{\sum (x - \bar{x})^2}{\sum f}} = \sqrt{\frac{252.8}{20}} = 3.56$$

$$\therefore \text{Skewness} = \frac{26.25 - 27.5}{3.56} = \underline{\underline{-0.35}} (D)$$

$$\begin{aligned}
 68) \text{ mode} &= L + \left[\frac{D_1}{D_1 + D_2} \right] \times C \\
 &= 14.5 + \left[\frac{(20-6)}{(20-6) + (20-15)} \right] \times 10 \\
 &= \underline{21.88} \quad \{ \text{No Answer} \}
 \end{aligned}$$

$$69) \bar{x} = \frac{\sum x_i}{n} = \frac{4+6+2+8+10+3+6+4+7+10}{10} = \frac{60}{10}$$

$$= 6$$

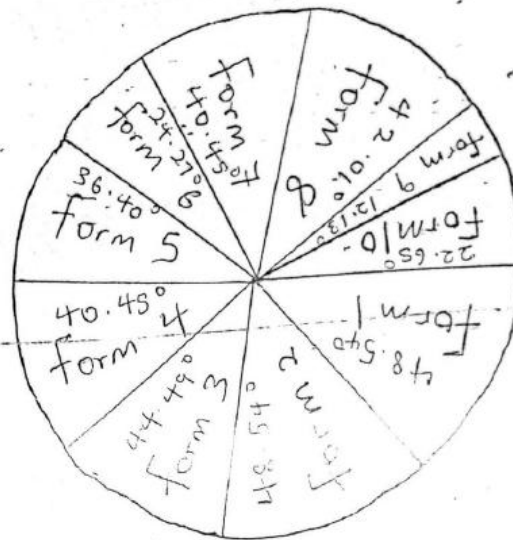
$$\begin{aligned}
 \text{Sum of Deviation} &= (4-6) + (6-6) + (2-6) + (8-6) + \\
 &\quad (10-6) + (3-6) + (6-6) + (4-6) + \\
 &\quad (7-6) + (10-6)
 \end{aligned}$$

$$\begin{aligned}
 &= -2 + 0 - 4 + 2 + 4 - 3 + 0 \\
 &\quad -2 + 1 + 4
 \end{aligned}$$

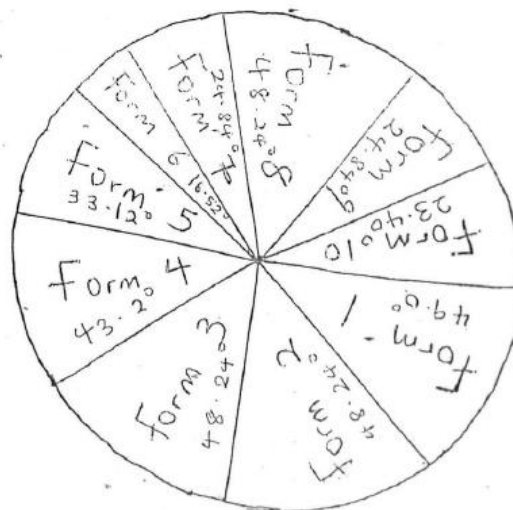
$$= \underline{0} \quad (C)$$

70) (A)

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Boys



Girls

(iii)

$$\text{Median} = \frac{n+1}{2}$$

for Boys $n = 445$

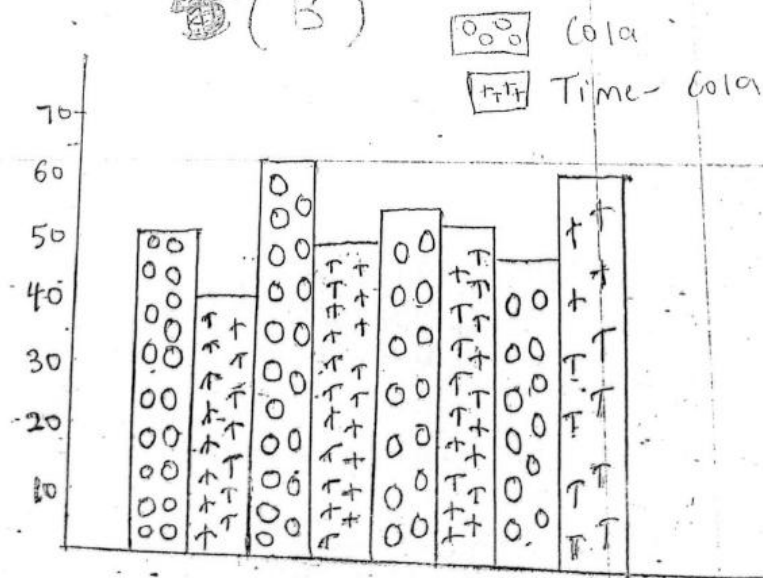
$$= \frac{445+1}{2} = \frac{446}{2} = 223$$

falls within Form 4 Boysfor Girls $n = 434$

$$= \frac{434+1}{2} = 217.5$$

falls within Form 4 Girls

(B)



ANSWER TO QUESTION 4

CLASS INTERVAL	CLASS BOUNDARY	MID CLASS (X)	frequency (f)	Cumulative freq. (cf)
56 - 58	55.5 - 58.5	57	7	7
59 - 61	58.5 - 61.5	60	13	20
62 - 64	61.5 - 64.5	63	68	88
65 - 67	64.5 - 67.5	66	144	232
68 - 70	67.5 - 70.5	69	197	429
71 - 73	70.5 - 73.5	72	204	633
74 - 76	73.5 - 76.5	75	208	841
77 - 79	76.5 - 79.5	78	160	1001
80 - 82	79.5 - 82.5	81	101	1102
83 - 85	82.5 - 85.5	84	54	1156
86 - 88	85.5 - 88.5	87	25	1181
89 - 91	88.5 - 91.5	90	13	1194
92 - 94	91.5 - 94.5	93	4	1198
95 - 97	94.5 - 97.5	96	2	1200

$$4(i) \quad \text{MEDIAN} = L_1 + \left[\frac{\frac{N}{2} - cf_b}{f_m} \right] \cdot C$$

Where Class Interval of the middle frequency $\Rightarrow 71-73$.

$$\therefore L_1 = 70.5 \quad N = 1200 \quad cf_b = 429$$

$$f_m = 204 \quad \text{Class width } (C) = 3$$

Paper 3 of 4 — MTH 102 PQ

Level 100 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS/COMPUTER SCIENCE

SECOND SEMESTER EXAMINATION 2019/2020

Course code & title: MTH 121 -General Mathematics II Time: 2HRS 15MINS

PAST QUESTIONS

- If $f(x) = \sqrt{x} - 2 + 3$ what is the $f(f(x)) \cap f(f(f(x)))$? (a) $(0, \infty)$ (b) $[2, \infty)$ (c) $(-1, \infty)$ (d) $[3, \infty)$
- Find the common domain for $f(x) = 4 + x$ and $g(x) = 2x + 1$? (a) 0 (b) 1 (c) 2 (d) 3
- Determine the domain and range of the following function $y = \sqrt{-4x} + 2$. (a) $x < 2, y \leq 0$ (b) $x \geq 2, y \leq 0$ (c) $x \leq 0, y \leq 0$ (d) $x \leq 0, y \geq 0$
- Find the maximum value of $y = -x^2 + 4$ for all values of x ? (a) 1 (b) 2 (c) ∞ (d) 4
- Suppose $f(x) = 2$, what is the domain of $f(x)$? (a) $(-\infty, -1) \cup (2, \infty)$ (b) $(-\infty, -1] \cup [2, \infty)$ (c) $[-\infty, -2]$ (d) $(-\infty, -1) \cup [2, \infty)$
- Suppose $f(x) = x^3 - 5x - 1$, find the limit as $x \rightarrow -3$ (a) 15 (b) -13 (c) -41 (d) -27
- Find $\lim_{x \rightarrow 9} \frac{x^2 - 9}{x - 9}$ (a) 2 (b) 0 (c) 6 (d) undefined
- Compute $\lim_{x \rightarrow 3} \frac{x^2 - 3}{x - 3}$ (a) 0 (b) 1 (c) -1 (d) indeterminate
- Find the limit of $\frac{1 - \cos x}{x^2}$ as x tends to 0 (a) 0 (b) 1 (c) 2 (d) $-\frac{1}{2}$
- Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ (a) 2 (b) 4 (c) 0 (d) indeterminate
- Which of the following is true for the function $f(x)$ defined by $f(x) = \begin{cases} x + 1 & x < -1 \\ -1 & -1 \leq x \leq 1 \\ x + 1 & x > 1 \end{cases}$ (a) f is continuous everywhere (b) f is continuous everywhere except at $x = -1, 1$ (c) f is continuous

everywhere except at $x = -1$ (d) f is continuous everywhere except at $x = 1$

$$x^2 + 2x \leq x$$

12. Find the values of k such that the function $f(x) = \begin{cases} \dots \end{cases}$ is continuous everywhere

$$-1 \leq x \leq 1$$

$$(a) f(x) = -1 \text{ for } x < -1 \quad (b) f(x) = -2 \text{ for } -1 < x < 1 \quad (c) f(x) = -1 \text{ for } x > 1 \quad (d) f(x) = -2 \text{ for } x > 1$$

$$x + x^2 \leq 2$$

13. Suppose $f(x) = \begin{cases} \dots \end{cases}$. Find a value for m such that the function is continuous at the point

$$1 + x^2 \leq 2$$

$$x = 2. \quad (a) m = 8 \quad (b) m = 1 \quad (c) m = 3 \quad (d) m = 2$$

$$x^4 + x^2 + 13$$

14. Find the limit $\lim_{x \rightarrow \infty} x^3 + 8x + 9$ (a) 0 (b) 1 (c) 2 (d) ∞

$$x \rightarrow \infty$$

$$x^2 + x + 1 \leq 1$$

15. The limit of $f(x) = (3x+2)^2$ as $x \rightarrow \infty$ is-- (a) 1 (b) 3 (c) 0 (d) 9

16. Suppose the total cost $C(x)$ of producing a quantity x of a product is given by the equation $C(x) = 500 +$

5x. The average cost per unit quantity $A(x)$ equals the total cost divided by the quantity produced x . Find

$$\lim_{x \rightarrow \infty} A(x). \quad (a) 5 \quad (b) 6 \quad (c) 5000 \quad (d) 5006$$

$$x \rightarrow \infty$$

$$-5x + 7 \leq x < 3$$

17. Let $f(x) = \begin{cases} \dots \end{cases}$. Find $\lim_{x \rightarrow 2^-} f(x)$ (a) 6 (b) -6 (c) -7 (d) -8

$$x - 16 \leq x \leq 3 \text{ for } x \rightarrow 3$$

$$2 \leq x \leq 3$$

18. If $\lim_{x \rightarrow 3} x + 2x + 1 = 16$. Find $x > 0$. (a) $x = \min \{1, \dots\}$ (b) $x = \max \{1, \dots\}$ (c) $x = \min \{1, \dots\}$ (d)

$$x \rightarrow 3 \quad 9 \leq x \leq 2$$

$$x$$

$$x = \max \{1, 12\}$$

$$1 \leq x$$

19. Suppose $x = 2$ and the limit of $1 + x^2$ is 5 as $x \rightarrow -2$. Find the minimum value of x (a) $x = 1$ (b) $x = 10$

$$1$$

$$(c) x = 2 \quad (d) 0$$

$$(h+4)^2 - 16$$

20. Compute $\lim_{h \rightarrow 0} h$

$$(a) \text{ indeterminate} \quad (b) 4 \quad (c) 6 \quad (d) 8$$

$$h \rightarrow 0$$

$$5 \leq x \leq 15 \quad -20 \leq x \leq 5$$

21. Find the derivative of $f(x) = 3x^4$ (a) 3 (b) 3 (c) $12x^3$ (d) $4x^3$

$$\sqrt{x} \sqrt{x^4} \leq 3 \sqrt{x^7} \sqrt{x}$$

$$3$$

22. Find the differential coefficient of $f(x) = 3\cos^4 x$ (a) $12\cos^4 x$ (b) $3\cos^4 x$ (c) $\cos^4 x$ (d) $\sin^4 x$

4

 $2^2 2^2 6^{-6}$ 23. Determine the derivative of $\sin(x) = \sin(x)$ (a) $\sin(x)$ (b) $3\sin(x)$ (c) $\sin(x)$ (d) $\sin(x)$ 24. Differentiate $(2x + 5)^2 (x - 4)$ (a) $4(2x + 5)(x - 4)$ (b) $4(2x + 5)(4x - 3)$ (c) $(2x + 5)(2x - 13)$ (d) $(2x + 5)(2x - 11)$ $\sin(x)$ 25. If $\sin(x) = 2x - 3$, $y = 3$ when $x = 0$, find y in terms of x . (a) $2x^2 - 3x$ (b) $x^2 - 3x$ (c) $2x^2 - 3x - 3$ (d) $2x^2 -$ $3x + 3$ 26. Find the derivative of $\sin(x) + \cos(x)^2(5x)$. (a) $10\sin(x)\cos(x)^2(5x)$ (b) $5\sin(x)\cos(x)^2(5x)$ (c) $2\sin(x)\cos(x)^2(5x)$ (d) $15\sin(x)\cos(x)^2(5x)$ 27. Find the derivative of the function $y = 2x^2(2x - 1)$ at the point $x = 1$ (a) 18 (b) 16 (c) -4 (d) -6 $x^7 - x^5$ 28. Find the derivative of $y = x^4$. (a) $x(x^2 - 1)$ (b) $3x(x^2 - 1)$ (c) $3x^2 - 1$ (d) $7x^6 - 5x^4$ 29. The derivative of $\csc x$ is ? (a) $\tan x \csc x$ (b) $-\cot x \csc x$ (c) $\tan x \sec x$ (d) $-\cot x \sec x$ 30. The distance covered by a particle is given by $s = 2t^2 - 4t^2 + 9$, find its acceleration when $t = 2$. (a)

2

 $22t/t^2$ (b) $30t/t^2$ (c) $12t/t^2$ (d) $9t/t^2$ 31. Differentiate $12x^3$. (a) $3x^4$ (b) $36x^2$ (c) $4x^3$ (d) $48x^3$

5

3

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5

3 6

33. Differentiate $6 \ln 2x$. (a) $6 \ln 2$ (b) $\ln x$ (c) $2x$ (d) x 34. Find the differential coefficient of $y = 3x^2 \sin 2x$. (a) $6x \sin 2x$ (b) $6x^2 \cos 2x$ (c) $6x(x \cos 2x + \sin 2x)$ (d) $-6x^2 \cos 2x$

-4 8

35. Differentiate $2x$. (a) 3 (b) $36x^2$ (c) $4x^3$ (d) $48x^3$ x^2

3 3 3 3

36. Differentiate $3\sqrt{x^2}$ (a) $9\sqrt{x^5}$ (b) $5\sqrt{x^2}$ (c) $15\sqrt{x^2}$ (d) $10\sqrt{x}$

2

37. Differentiate $y = 2 \sin 3x$. (a) $2 \cos 3x$ (b) $-2 \cos 3x$ (c) $6 \cos 3x$ (d) $3 \cos 3x$

3 3 15 3 15

38. Differentiate $5x$ (a) - (b) $5x$ (c) (d) $-5x$ $\sin 5x \sin 5x \sin 5x \sin 5x$ 4 $\sin 5x$

39. Find the differential coefficient of $y = 5x^4$

4

(a) $5(5x^4 \cos 5x - 4 \sin 5x)$ (b) $5x^4 \cos 5x - 4 \sin 5x$ (c) $5x^4 \cos 5x$ (d) $-4 \sin 5x$

5x

40. Differentiate $y = 3 \cos(5x^2 + 2)$. (a) $-15x \sin(5x^2 + 2)$ (b) $-15x \sin 5x^2$ (c) $-3 \sin(5x^2 + 2)$

(d) $-30x \sin(5x^2 + 2)$

41. The length y meters of a certain metal rod at temperature x° is given by $y = 1 + 0.00005x + 0.0000004x^2$. Determine the rate of change of length in $\text{mm}/^\circ$ when the temperature is 100° .

(a) $0.00013 \text{ mm}/^\circ$ (b) $0.013 \text{ mm}/^\circ$ (c) $0.05 \text{ mm}/^\circ$ (d) $0.00005 \text{ mm}/^\circ$

42. The luminous intensity y candelas of a lam at varying voltage V is given by $y = 4 \times 10^{-4} V^2$. Determine the

voltage at which light is increasing at a rate 0.6 candelas per volt. (a) 1500 V (b) 500 V (c) 750 V (d) 1000 V

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the acceleration when $x = 1.5$. (a) 4 m/s^2 (b) 18.25 m/s^2 (c) 24 m/s^2 (d) 23 m/s^2

44. If the differential function is given as $f(x) = 6x^2 + 7$, what is the integral of the function? (a) $3x^2 + 7x + C$

(b) $3x^3 + 7x^2 + C$ (c) $6x^2 + 7x + C$ (d) $x^2 - 7x + C$

1

45. Evaluate the integral of $f(x) = x^2 - 2x$ (a) $\frac{1}{3}x^3 - x^2 + C$ (b) $-\frac{1}{3}x^3 - x^2 + C$ (c) $\frac{1}{3}x^3 + C$ (d) $\frac{1}{3}x^3 - x^2 + C$

1 1 1 1 1

46. Evaluate the integral $\int_0^1 x^2 dx$. (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2} + \frac{1}{3}$

1 1 1

47. Evaluate the integral $\int_0^1 3x^2 dx$. (a) $x^3 + C$ (b) $-\frac{1}{3}x^3 + C$ (c) $\frac{1}{3}x^3 + C$ (d) $\frac{1}{3}x^3$

2 2 1 2

48. Evaluate $\int x^{1/2} dx$ (a) $\frac{2}{3}x^{3/2} + C$ (b) $\frac{2}{3}x^2 + C$ (c) $\frac{2}{3}x^{-1/2} + C$ (d) $\frac{2}{3}x^{3/2}$

3 3 2 3

1 1 1

49. Evaluate $\int_0^1 7x^6 dx$ (a) $\frac{7}{7}x^7 + C$ (b) $-\frac{7}{7}x^7 + C$ (c) $\frac{7}{7}x^7 + C$ (d) $-\frac{7}{7}x^7$

50. Evaluate $\int_0^1 x^2 dx$. (a) $\frac{1}{3}x^3 - \frac{1}{3}x^3 + C$ (b) $\frac{1}{3}x^3 + \frac{1}{3}x^3 + C$ (c) $\frac{1}{3}x^3 + \frac{1}{3}x^3 + C$ (d) $x + c$

2 2 1 2

51. Find $\int_0^1 x^2 dx$. (a) $\frac{2}{3}x^3 - 4 + C$ (b) $\frac{2}{3}x^3 + C$ (c) $\frac{2}{3}x^3 + C$ (d) $\ln x$

1

52. Find $\int_0^1 x^2 dx$. (a) $\frac{2}{3}(x^3 - x^3) + C$ (b) $\frac{2}{3}x^3 + C$ (c) $\frac{2}{3}x^3 + C$ (d) $\cos x$

1 1 1 1

53. Find $\int_0^1 3x^2 dx$. (a) $-\frac{3}{2}(x^2 + 2x^2) + C$ (b) $-\frac{3}{2}x^2 + C$ (c) $-\frac{3}{2}x^2 + C$ (d) $-\frac{3}{2}x^2$

2 2 2 2

2 2 2 2

54. Find the integral of $\int_0^2 (x^2 - 4) dx$ (a). $2 - 4$

+ (b). $2 - 4$

+ (c). $4 - 2$

+ (d). $2 + 4$

+ (e).

$\int_0^2 (x^2 - 4) dx = \int_0^2 x^2 dx - \int_0^2 4 dx$

55. Evaluate $\int_0^2 (x^2 - 4) dx$ (a). $2 - 4$

+ (b). $2 - 4$

+ (c). $4 - 2$

+ (d). $2 + 4$

+ (e).

1 1 1 1 1

56. Integrate $\int_0^2 \sin 4x \cos 2x dx$ (a). $-\frac{1}{6} (\cos 6x + \cos 2x) + \frac{1}{6}$ (b). $-\frac{1}{6} (\cos 6x + \sin 2x) + \frac{1}{6}$ (c).

$\frac{2}{6} \sin 6x + \frac{2}{6} \cos 2x$

1 1 1 1 1

– $\frac{2}{6} (\sin 6x + 2 \cos 2x) + \frac{1}{6}$ (d). $-\frac{2}{6} (2 \cos 6x + 6 \cos 2x) + \frac{1}{6}$

1 1 1 1 1

57. Evaluate $\int \cos 6x \cos 4x dx$ (a). $-\frac{2}{10} (\cos 10x + 2 \cos 2x) + \frac{1}{10}$ (b). $\frac{2}{10} (\sin 10x + 2 \sin 2x) + \frac{1}{10}$ (c).

1 1 1 1 1

– $(\sin 10x + \cos 2x) + \frac{1}{10}$ (d). $(\cos 10x + \sin 2x) + \frac{1}{10}$

$\frac{2}{10} \sin 2x + \frac{2}{10} \cos 10x$

$\frac{1}{10} \sin 2x + \frac{1}{10} \cos 10x$

58. Find the integral of $\int_0^2 (x^2 - 4) dx$ (a). $\frac{1}{3} x^3 - 4x + \frac{4}{3}$ (b). $\frac{1}{3} x^3 - 4x + \frac{4}{3}$ (c). $-\frac{1}{3} x^3 - 4x + \frac{4}{3}$ (d).

$\frac{1}{3} x^3 - 4x + \frac{4}{3}$

1 1

– $\frac{2}{3} x^3 - 4x + \frac{4}{3}$

59. Given the integrand $\sqrt{x^2 - 4}$, what's the substitution for x ? (a). $\sqrt{x^2 - 4}$ (b). $\sqrt{x^2 - 4}$ (c).

$\sqrt{x^2 - 4}$ (d).

$\sqrt{x^2 - 4}$

60. What's the substitution for x in the integrand $\sqrt{x^2 - 4}$? (a). $\sqrt{x^2 - 4}$ (b). $\sqrt{x^2 - 4}$ (c). $\sqrt{x^2 - 4}$ (d).

$\sqrt{x^2 - 4}$

61. What's the substitution for x in the integrand $\sqrt{x^2 + 4}$? (a). $\sqrt{x^2 + 4}$ (b). $\sqrt{x^2 + 4}$ (c). $\sqrt{x^2 + 4}$ (d).

$\sqrt{x^2 + 4}$

$\sqrt{x^2 + 4}$

62. If $y = 2x^3 + 9x^3 - 3x^2$, then $\frac{dy}{dx} =$ (a) $9x^3 + 9x^3 - 6x$ (b) $18x^3 - 6x$ (c) $6x^3 + 18x^3 - 18x$

$18x^3 - 6x$

63. If $y = \tan(x^2 - x^2)$, is (a) $\frac{dy}{dx} = 2x \sec^2(x^2 - x^2)$ (b) $2y \sec^2(x^2 - x^2)$ (c) $\sec^2(x^2 - x^2)$ (d) $y \sec^2(x^2 - x^2)$

$\sec^2(x^2 - x^2)$

$\frac{dy}{dx} = 2x \sec^2(x^2 - x^2)$

64. If $y = x^2 - 3x$, then $\frac{dy}{dx} =$ (a) $9x^3 - 2x$ (b) $9x^2 - 3x$ (c) $x^3 - 2x$ (d) $3x^2 - 3x$

x^2

$x + x, x^2 - 2x - 2x - 2x - 2x - 2x + 2x - 2x + 2x$

65. If $f(x) = x^3 - 3x^2 + 2x$, then $f'(x)$ is (a) $(x-1)^3$ (b) $(x-1)^2$ (c)

$(x-1)^3$

(d)

$(x-1)^2$

$f'(x) = 3x^2 - 6x + 2$

66. If $\sin x + \cos x = 2$, then $\sin x \cdot \cos x$ is (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d)

$\frac{1}{2} + \cos x + \sin x = 2$

67. If the second derivative of a function at point x is positive, it implies that the function is (a) concave downward (b) concave upward (c) hyperbola (d) None of the above

5 5 5

5 5 5

68. Given that $f(x) = 2x^3 - 5x^2 + 4$, determine the point of inflection (a) 6 (b) 3 (c) 0 (d) -6

69. $f(x) = x^3 - 3x$, the point of inflection for the curve is (a) 6 (b) -1 (c) 0 (d) -6

70. An inflection point of the curve $f(x) = x^3 - 6x^2 + 9x$ is (a) 2 (b) -6 (c) 6 (d) -2

Paper 4 of 4 — MTH 102

Level 100 • Second Semester • Handout

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS/COMPUTER SCIENCE

SECOND SEMESTER EXAMINATION 2019/2020

Course code & title: MTH 121 -General Mathematics II Time: 2HRS 15MINS

PAST QUESTIONS

- If $f(x) = \sqrt{x} - 2 + 3$ what is the domain of $f(x)$? (a) $(0, \infty)$ (b) $[2, \infty)$ (c) $(-1, \infty)$ (d) $[3, \infty)$
- Find the common domain for $f(x) = 4 + x$ and $g(x) = 2x + 1$? (a) 0 (b) 1 (c) 2 (d) 3
- Determine the domain and range of the following function $f(x) = \sqrt{-4x} + 2$. (a) $x < 2, y \leq 0$ (b) $x \geq 2, y \leq 0$ (c) $x \leq 0, y \leq 0$ (d) $x \leq 0, y \geq 0$
- Find the maximum value of $f(x) = -x^2 + 4x + 4$ for all values of x ? (a) 1 (b) 2 (c) ∞ (d) 4
- Suppose $f(x) = 2x - 2$, what is the domain of $f(x)$? (a) $(-\infty, -1) \cup (2, \infty)$ (b) $(-\infty, -1] \cup [2, \infty)$ (c) $(-\infty, -1) \cup [2, \infty)$ (d) $(-\infty, -1] \cup (2, \infty)$
- Suppose $f(x) = x^3 - 5x - 1$, find the limit as $x \rightarrow -3$ (a) 15 (b) -13 (c) -41 (d) -27
- Find $\lim_{x \rightarrow 9} \frac{x^2 - 9}{x - 9}$ (a) 2 (b) 0 (c) 6 (d) undefined
- Compute $\lim_{x \rightarrow 3} \frac{x^2 - 3x}{x^2 - 1}$ (a) 0 (b) 1 (c) -1 (d) indeterminate
- Find the limit of $\frac{1 - \cos x}{x^2}$ as x tends to 0 (a) 1 (b) 2 (c) 0 (d) $-\frac{1}{2}$
- Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ (a) 2 (b) 4 (c) 0 (d) indeterminate
- Which of the following is true for the function $f(x)$ defined by $f(x) = \begin{cases} x + 1 & x < -1 \\ x - 1 & -1 \leq x \leq 1 \\ x + 1 & x > 1 \end{cases}$ (a) f is continuous everywhere (b) f is continuous everywhere except at $x = -1, 1$ (c) f is continuous

everywhere except at $x = -1$ (d) f is continuous everywhere except at $x = 1$

$$x^2 + 2x \leq x$$

12. Find the values of k such that the function $f(x) = \begin{cases} \dots \end{cases}$ is continuous everywhere

$$-1 \leq x \leq 1$$

$$(a) f(x) = -1 \text{ for } x < -1 \quad (b) f(x) = -2 \text{ for } x < -1 \quad (c) f(x) = -1 \text{ for } x < 1 \quad (d) f(x) = -2 \text{ for } x < 2$$

$$x + x^2 \leq 2$$

13. Suppose $f(x) = \begin{cases} \dots \end{cases}$. Find a value for m such that the function is continuous at the point

$$1 + x^2 \leq 2$$

$$x = 2. \quad (a) m = 8 \quad (b) m = 1 \quad (c) m = 3 \quad (d) m = 2$$

$$x^4 + x^2 + 13$$

14. Find the limit $\lim_{x \rightarrow \infty} x^3 + 8x + 9$ (a) 0 (b) 1 (c) 2 (d) ∞

$$x \rightarrow \infty$$

$$x^2 + x + 1 \leq 1$$

15. The limit of $f(x) = (3x+2)^2$ as $x \rightarrow \infty$ is-- (a) 1 (b) 3 (c) 0 (d) 9

16. Suppose the total cost $C(x)$ of producing a quantity x of a product is given by the equation $C(x) = 500 +$

5 x . The average cost per unit quantity $A(x)$ equals the total cost divided by the quantity produced x . Find

$$\lim_{x \rightarrow \infty} A(x). \quad (a) 5 \quad (b) 6 \quad (c) 5000 \quad (d) 5006$$

$$x \rightarrow \infty$$

$$-5x + 7 \leq x < 3$$

17. Let $f(x) = \begin{cases} \dots \end{cases}$. Find $\lim_{x \rightarrow 2^-} f(x)$ (a) 6 (b) -6 (c) -7 (d) -8

$$x - 16 \leq x \leq 3 \text{ for } x \rightarrow 3$$

$$2 \leq x \leq 3$$

18. If $\lim_{x \rightarrow 3} x + 2x + 1 = 16$. Find $x > 0$. (a) $x = \min \{1, \dots\}$ (b) $x = \max \{1, \dots\}$ (c) $x = \min \{1, \dots\}$ (d)

$$x \rightarrow 3 \quad 9 \leq x \leq 2$$

$$x$$

$$x = \max \{1, 12\}$$

$$1 \leq x$$

19. Suppose $x = 2$ and the limit of $1 + x^2$ is 5 as $x \rightarrow -2$. Find the minimum value of x (a) $x = 1$ (b) $x = 10$

$$1$$

$$(c) x = 2 \quad (d) 0$$

$$(h+4)^2 - 16$$

20. Compute $\lim_{h \rightarrow 0} h$

$$(a) \text{ indeterminate} \quad (b) 4 \quad (c) 6 \quad (d) 8$$

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$$5 \leq x \leq 15 \quad -20 \leq x \leq 5$$

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$$\sqrt{x} \sqrt{x^4} \leq 3 \sqrt{x^7} \sqrt{x}$$

$$3$$

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4

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2

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5

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3 3 3 3

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2

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3 3 15 3 15

38. Differentiate $5x$ (a) $-$ (b) $5x$ (c) $(d) -5x$ $\sin 5x \sin 5x \sin 5x \sin 5x$ $4 \sin 5x$

39. Find the differential coefficient of $y = 5x^4$

4

(a) $5(5x^4 \cos 5x - 4 \sin 5x)$ (b) $5x^4 \cos 5x - 4 \sin 5x$ (c) $5x^4 \cos 5x$ (d) $-4 \sin 5x$

$5x^4$

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(b) $3x^3 + 7x^2 + C$ (c) $6x^2 + 7x + C$ (d) $x^2 - 7x + C$

1

45. Evaluate the integral of $f(x) = x^2 - 2x$ (a) $\frac{1}{3}x^3 - x^2 + C$ (b) $-\frac{1}{3}x^3 - x^2 + C$ (c) $\frac{1}{3}x^3 + C$ (d) $\frac{1}{3}x^3 - x^2 + C$

1 1 1 1 1

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1 1 1

47. Evaluate the integral $\int_0^1 3x^2 dx$. (a) $x^3 + \frac{1}{3}$ (b) $-\frac{1}{3}x^3 + \frac{1}{3}$ (c) $\frac{1}{3}x^3 + \frac{1}{3}$ (d) $\frac{1}{3}x^3 - \frac{1}{3}$

2 2 1 2

48. Evaluate $\int x^{1/2} dx$ (a) $\frac{2}{3}x^{3/2} + C$ (b) $\frac{2}{3}x^2 + C$ (c) $\frac{2}{3}x^{-1/2} + C$ (d) $\frac{2}{3}x^{3/2}$

3 3 2 3

1 1 1

49. Evaluate $\int_0^1 7x^2 dx$ (a) $\frac{7}{3}x^3 + \frac{1}{3}$ (b) $-\frac{7}{3}x^3 + \frac{1}{3}$ (c) $\frac{7}{3}x^3 + \frac{1}{3}$ (d) $-\frac{7}{3}x^3 + \frac{1}{3}$

1 1 1 1

50. Evaluate $\int_0^1 x^2 dx$. (a) $\frac{1}{3}x^3 - \frac{1}{3}x^2 + \frac{1}{3}$ (b) $\frac{1}{3}x^3 + \frac{1}{3}x^2 + \frac{1}{3}$ (c) $\frac{1}{3}x^3 + \frac{1}{3}x^2 + \frac{1}{3}$ (d) $x + c$

2 2 1

51. Find $\int_0^1 x^2 dx$. (a) $\frac{2}{3}x^3 - 4 + \frac{1}{3}$ (b) $\frac{1}{3}x^3 + \frac{1}{3}$ (c) $\frac{1}{3}x^3 + \frac{1}{3}$ (d) $\ln x$

1

52. Find $\int_0^1 x \cos x dx$. (a) $2(\frac{1}{2}x^2 \cos x - \frac{1}{2}x \sin x) + \frac{1}{2}$ (b) $\frac{1}{2}x \cos x + \frac{1}{2}$ (c) $\frac{1}{2}x \cos x + \frac{1}{2}x \sin x + \frac{1}{2}$ (d) $\cos x$

1 1 1 1

53. Find $\int_0^1 3x^2 dx$. (a) $-\frac{3}{2}(\frac{1}{2}x^2 \cos x + 2 \sin x) + \frac{1}{2}$ (b) $-\frac{3}{2}x^2 \cos x + \frac{1}{2}$ (c) $-\frac{3}{2}x^2 \cos x + \frac{1}{2}$ (d) $-\frac{3}{2}x^2 \cos x$

$\frac{1}{2}x^2 \cos x - \frac{1}{2}x \sin x + \frac{1}{2}$

54. Find the integral of $\int_0^2 (x^2 - 4) dx$ (a). $2 - 4$

+ (b). $2 + 4$

+ (c). $4 - 2$

+ (d). $2 + 4$

+ (e). $2 - 4$

55. Evaluate $\int_0^2 (x^2 - 4) dx$ (a). $2 - 4$

+ (b). $2 + 4$

+ (c). $4 - 2$

+ (d). $2 + 4$

+ (e). $2 - 4$

56. Integrate $\int_0^2 (x^2 - 4) dx$ (a). $-(\cos 6x + \cos 2x) + C$

+ (b). $-(\cos 6x + \sin 2x) + C$

+ (c). $2(6 \sin 6x + 2 \cos 2x) + C$

+ (d). $-2(2 \cos 6x + 6 \cos 2x) + C$

+ (e). $-2(2 \cos 6x + 6 \cos 2x) + C$

57. Evaluate $\int \cos 6x \cos 4x dx$ (a). $-2(10 \cos 10x + 2 \cos 2x) + C$

+ (b). $2(10 \sin 10x + 2 \sin 2x) + C$

+ (c). $-(\sin 10x + \cos 2x) + C$

+ (d). $(\cos 10x + \sin 2x) + C$

58. Find the integral of $\int_0^2 (x^2 - 4) dx$ (a). $\frac{1}{2}x^2 - 4x + C$

+ (b). $\frac{1}{2}x^2 - 4x + C$

+ (c). $-\frac{1}{2}x^2 - 4x + C$

+ (d). $\frac{1}{2}x^2 + 4x + C$

59. Given the integrand $\sqrt{x^2 - 4}$, what's the substitution for x (a). $u = x^2 - 4$

+ (b). $u = x^2 + 4$

+ (c). $u = x^2 - 4$

+ (d). $u = x^2 + 4$

60. What's the substitution for x in the integrand $\sqrt{x^2 - 4}$ (a). $u = x^2 - 4$

+ (b). $u = x^2 + 4$

+ (c). $u = x^2 - 4$

+ (d). $u = x^2 + 4$

61. What's the substitution for x in the integrand $\sqrt{x^2 + 4}$ (a). $u = x^2 + 4$

+ (b). $u = x^2 - 4$

+ (c). $u = x^2 + 4$

+ (d). $u = x^2 - 4$

62. If $y = 2x^3 + 9x^2 - 3x - 2$, then $\frac{dy}{dx} =$ (a). $9x^2 + 9x - 3$

+ (b). $9x^2 + 9x - 3$

+ (c). $9x^2 + 9x - 3$

+ (d). $9x^2 + 9x - 3$

x^2

$x + x, x^2 - 2x - 2x - 2x - 2x - 2x + 2x - 2x + 2x$

65. If $f(x) = x^3 - 3x^2 + 2x$, then $f'(x)$ is (a) $(x-1)^3$ (b) $(x-1)^2$ (c)

$(x-1)^3$

(d)

$(x-1)^2$

$f'(x) = 3x^2 - 6x + 2$

66. If $\sin x + \cos x = 2$, then $\sin x \cdot \cos x$ is (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d)

$\sin x + \cos x = 2 \Rightarrow \sin x = \cos x = 1$

67. If the second derivative of a function at point x is positive, it implies that the function is (a) concave downward (b) concave upward (c) hyperbola (d) None of the above

5 5 5

68. Given that $f(x) = 2x^3 - 5x^2 + 4$, determine the point of inflection (a) 6 (b) 3 (c) 0 (d) -6

69. $f(x) = x^3 - 3x$, the point of inflection for the curve is (a) 6 (b) -1 (c) 0 (d) -6

70. An inflection point of the curve $f(x) = x^3 - 6x^2 + 9x$ is (a) 2 (b) -6 (c) 6 (d) -2